USING AI TOOLS TO INVESTIGATE THE HEALTH EFFECTS OF TRAFFIC RELATED $PM_{2.5}$ AIR POLLUTION ON CHRONIC DISEASE PREVALENCE IN THE GEORGETOWN COMMUNITY OF SALISBURY, MARYLAND

by

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ABSTRACT

Title of Thesis:	Using AI tools to investigate the health effects of traffic related $PM_{2.5}$ air pollution on chronic disease prevalence in the Georgetown community of Salisbury, Maryland
	Katrina V Kelly, Master of Science 2025
Thesis Directed By:	Joseph S. Pitula, Ph.D., Director of Research, Office of Sponsored Research and Programs (OSRP)

Research has established that $PM_{2.5}$ is a major public and environmental health concern and a key driver of cardiovascular disease, diabetes, and hypertension. $PM_{2.5}$ contains a variety of pollutants such as carbon dioxide (CO_2), lead (Pb), arsenic (As), nitrogen oxide (NO_x), among others that penetrate the air-blood barrier to internal organs, affecting gene expression, cell formation, and fetal development *in utero.* Although predominant medical theory espouses genetic causes of these chronic diseases, recent research has begun to consider cumulative and environmental exposures like poor air quality that contribute to epidemiological pathologies (Wild, 2012). These nonmedical factors are referred to as social determinants of health (SDOH) and fall under an emerging branch of 'omics' known as *Exposomics* (Neufcourt, et al, 2022). The purpose of this research is to use AI tools to investigate the environmental and non-medical relationships between prolonged exposure to vehicular $PM_{2.5}$ and chronic disease mortality (e.g., cardiovascular disease, diabetes, and hypertension) within the community of Georgetown located in Salisbury,

Maryland. For this analysis, AI algorithms will be used to extract geographic, demographic, and mortality data from digitized certificates of death (CODs) from the period between 1911 to 1922 (n = 1680) – obtained from the Maryland State Archives. Consistent with prevailing studies, we anticipate (1) that our results will evidence an upward trend in the incidence of chronic disease mortality related to fine particulate air pollution exposure given the proximity to major roadways and higher traffic volume driven by expansion of population and industry in the area, and (2) that AI will be a viable tool for data extraction and compilation. Additional value and insight could be gained from an artificial intelligence (AI) tool designed to collect and analyze pre- and post-highway construction disease prevalence in the selected study areas that predate data collection periods prior to 1970. Such analysis could have positive implications and adaptations across other disciplines to help inform policy and strategies related to ecotoxicology and human health.

DEDICATION

This graduate study and thesis are in memory of my parents – Castella Hubbard-Kelly and William Kelly, Jr. They believed firmly in the power of education and would have been pleased to see that the winding and nontraditional path I have followed for most of my life – often to their dismay - finally culminated in an advanced degree. I also dedicate this thesis to my extended family and friends for their loving encouragement and financial support which was always given in the right amount at just the right time. This academic journey has been one of endurance and faith as much as it was of learning and scientific discovery; I owe a profound debt gratitude to the Buddha Dharma to which I chanted *Nam Myoho Renge Kyo* day and night to achieve this milestone.

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ACKNOWLEDGEMENTS

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Many thanks are also expressed to the University of Maryland College Park and to the institution and legacy of the University of Maryland Eastern Shore. Go Hawks!

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CHAPTER 1: INTRODUCTION

Understanding the effects that the environment has on human health is a profoundly complex undertaking and with the events of the past five years – the COVID-19 pandemic, resurgence of social and environmental justice movements, and national and global agendas to address climate change and sustainability – scientists, researchers and health professionals have recognized that the current approach to environmental and public health management, especially chronic diseases, may not be sufficient to achieve the social and public health objectives that have been established.

Considering the numerous interdependent and interacting variables (e.g., family history, diet and lifestyle, geographic location, environmental contaminants, and even access to resources, etc.) a branch of 'omics' known as *Exposomics* has emerged to incorporate the environmental and the nonmedical factors, referred to as social determinants of health (SDOH), that can combine to result in either positive or negative health outcomes at the individual and sometimes community level.

Within an exposomic framework, this research considers environmental quality within the context of public health and data science to investigate chronic diseases linked to environmental factors. It is centered in the framework of the Coupled Human and Natural Systems (CHANS), a description of a dynamic system with interconnected social and environmental feedback loops and challenges in causal inference due to the difficulty in isolating environmental and human variables (Ferraro et al., 2019).

Highlighted in the work of Ferraro et al. are the challenges of causal inference within CHANS due to the interconnected nature of social and environmental dimensions. Their research underscores the necessity of structural knowledge and interdisciplinary collaboration to navigate the complexities of excludability and interference in causal analyses. The CHANS approach recognizes the bidirectional influence of human and natural systems, where ecological disturbances and human actions are mutually reinforcing, necessitating nuanced models to capture these dynamics. As Marina et al. (2011) elaborate. The CHANS builds on extensive scholarship concerning human–nature interactions, integrating diverse disciplinary insights to model coupled systems effectively.

For example, epidemiologists and sociologists emphasize the significance of integrating social factors into exposome research to comprehend health outcomes and inequalities accurately. Neglecting social aspects in environmental health studies could lead to underestimating exposure effects and overlooking disparities in exposure distribution and health outcomes.

Aligned with the inherently interdisciplinary mission of the Environment, Health and Society Foundation of the Marine, Estuarine, and Environmental Science (MEES) graduate program, this research project is informed by the complexities the CHANS, the Exposome and the evolving technologies that may be leveraged to improve analysis, interventions, and outcomes.

The Coupled Human and Natural System

The Exposome

The concept of the exposome indicates all forms of exposure that an individual may experience throughout the course of a lifetime which encompasses diet, physical, social, and occupational environments, lifestyle choices, as well as an individual's genetics and/or epigenetics, and physiology; the term *Exposomics* refers to the application of assessing all exposures - internal and external – of all *omics* disciplines (i.e., genomics, metabolomics, lipidomics, transcriptomics and proteomics). The Exposomic model for evaluating health risk considers personal attributes as well as social determinants like environmental and social stressors, economic status, diet, and lifestyle, as well as genetic factors and employs internal and external exposure assessment, with "omics" technologies (like metabonomics and adductomics) offering insight into disease causation. Continued research is required to validate their utility in defining exposome and Exposomic scientists are increasingly incorporating social determinants of health, such as gender, race, education, income, housing, and food security, into the framework, particularly within the external exposome for a more comprehensive analysis of disease outcomes.

Understanding how these interactions with genetics, physiology, and epigenetics influence health involves consideration of correlations as well as causal links between environmental exposures and diseases, aiming to identify crucial steps in the etiology of diseases. Researchers argue that studying causality is essential for validating biomarkers and understanding the complexity of chronic diseases like cancer, diabetes, or cardiovascular diseases, which may involve multiple weak and strong exposures. While some advocate focusing solely on metabonomics for exposome research, as small molecular compounds can signal biological regulation and uncover disease mechanisms, its exclusivity in articulating exposome aspects remains uncertain (NIESH, n.d.)

Social Determinants of Health

Numerous studies highlight that ethnic, minoritized and low-income communities are disproportionately exposed to the environmental risks posed by industrialization, militarization, and consumer practices, and these communities can meet with significant obstacles to prevent the establishment of harmful industries in their neighborhoods that can exacerbate existing disparities. These mentioned factors, along with other characteristics of a community such as income level, socioeconomic status, and educational attainment, are referred to as nonmedical or social determinants of health (SDOH) (Bona et al., 2022). They provide critical context for the concentration of chronic disease among minoritized groups and can contribute valuable insight for policymakers seeking to foster equity among community stakeholders (Mohai et al., 2018).

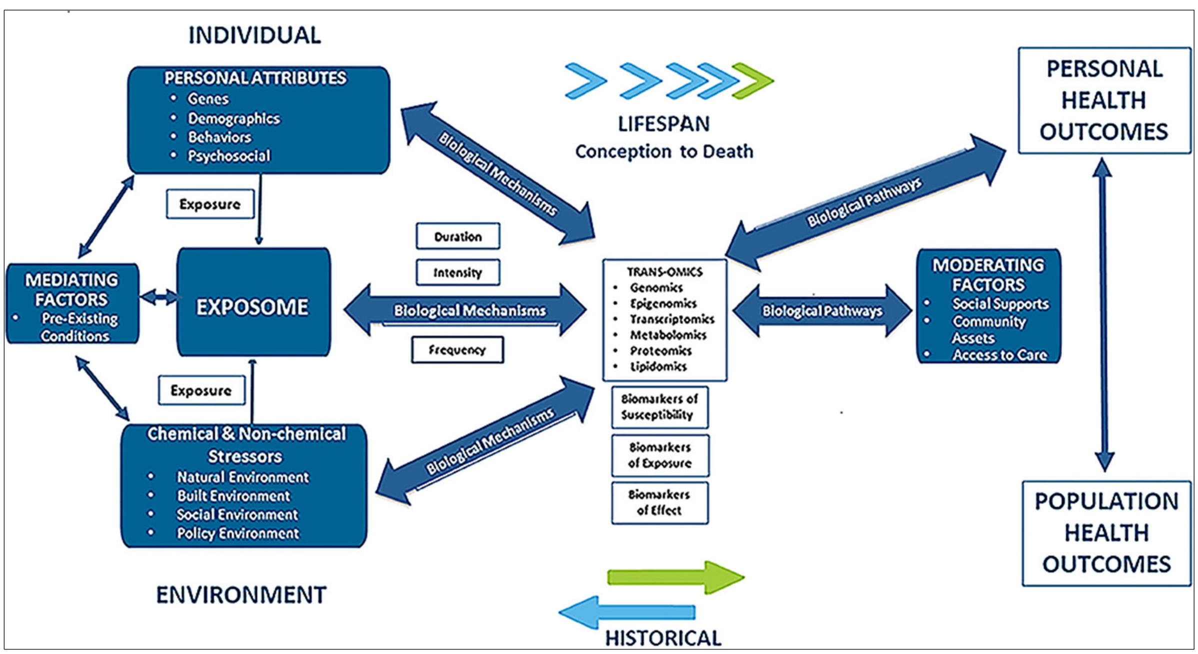

Figure 1 Exposome and Social Determinants of Health, Juarez, Frontier in public health (2020): 379

Pollutants and Public Policy

The Clean Air Act

The Clean Air Act (CAA), a landmark piece of U.S. environmental legislation first enacted in 1970 and amended in 1977 and 1990, serves as a comprehensive framework for regulating air pollutants to safeguard public health and the environment. Under the Act, the Environmental Protection Agency (EPA) is authorized to set National Ambient Air Quality Standards (NAAQS) and regulate hazardous air pollutants (HAPs). The Act's implementation includes state-level State Implementation Plans (SIPs) to meet air quality standards, though many regions required deadline extensions due to non-compliance. The 1990 amendments introduced technology-based standards for major and area sources of HAPs, periodic reviews for risk mitigation, and expanded enforcement mechanisms. These measures reflect the Act's evolution toward addressing complex air quality issues.

The 1990 amendments, signed into law by President George H.W. Bush, targeted four key areas: acid rain, urban air pollution, toxic emissions, and ozone layer depletion and led to innovative measures such as market-based solutions for sulfur dioxide reduction and enhanced enforcement mechanisms that significantly improved air quality. By the early 1990s, areas with previously unhealthy carbon monoxide levels achieved compliance with EPA standards noted in the NAAQS Table (EPA, n.d.) (APPENDIX B), while reductions in ozone and particulate pollution were attributed to cleaner fuels and advanced vehicle technologies. Additionally, the phase-out of ozone-depleting substances accelerated public health benefits. The EPA projected that by 2020, these amendments would prevent over 230,000 premature deaths annually and generate economic benefits approaching $2 trillion, demonstrating the Act's cost-effectiveness (EPA, 2015).

The CAA legislative history highlights a shift from regional strategies to a national regulatory framework, driven by public demand and environmental crises; its amendments exemplify adaptive policymaking in response to scientific advancements and societal needs, integrating environmental protection with economic considerations. Today, the Act remains a cornerstone of U.S. environmental policy, underscoring the ongoing commitment to reducing air pollution and addressing emerging environmental challenges.

The Federal-Aid Highway Act

The post-war period saw renewed momentum in the United States, driven by concerns about national defense, urban growth, and employment for returning soldiers. During this time, federal leadership, including contributions from the Bureau of Public Roads (BPR) and Public Roads Administration (PRA), collaborated with state and local agencies to develop plans and design standards for an integrated interstate system. Influenced by both practical needs and visionary projects like the 1939 New York World's Fair "Futurama" exhibit, the groundwork for modern freeways emerged. Spurred by an ambition to unite the country through advanced transportation infrastructure, the Federal-Aid Highway Act of 1956 emerged. This legislation laid the foundation for the modern Interstate Highway System, introducing a transformative 43,000-kilometer non-toll interregional highway network, emphasizing improved traffic flow, access control, and urban development (Weingroff, 1996). This vision, further refined in the 1943 *Interregional Highways* report, advocated for a 63,000-kilometer system designed for long-term traffic needs and urban planning; yet, political and funding disagreements delayed substantial progress, with the Federal-Aid Highway Act of 1944 merely authorizing, but not prioritizing or funding, a 65,000-kilometer interstate system. Although this network of national roadways symbolized President Eisenhower's belief in its critical role for a united nation, it introduced notable historical and environmental consequences that continue even today (Karas, 2015).

The development of the U.S. interstate highway system, particularly from the 1950s through the 1970s, was not merely a feat of engineering, but a process laden with racialized decision-making that disproportionately affected Black communities. Empirical research has demonstrated that the proximity of highways to urban neighborhoods significantly influenced demographic patterns. Mahajan's study (2024) shows that census tracts located closer to highways experienced a marked increase in the share of Black residents between 1970 and 1990, while nearby white populations declined, particularly in tracts where highway proximity was greatest. These shifts suggest that highway siting catalyzed racial sorting by facilitating white flight and concentrating Black residents in areas cut off or degraded by the infrastructure itself (Mahajan, 2024).

Although legislative attempts in the 1960s aimed to mitigate the destruction caused by such infrastructure—by requiring relocation housing and protecting parks and historic districts—these measures were insufficient. Archer (2020) argues that existing legal frameworks fail to prioritize racial equity or offer meaningful reinvestment in the communities most harmed by highway construction (Archer, 2020). The legacy of these decisions is visible in the physical and social landscapes of urban America, where Black neighborhoods were not only disrupted but left to languish in underinvestment and isolation. As such, the history of highway siting is not merely one of transportation policy, but one of racialized spatial reordering with enduring consequences.

Ultimately, the passage of the Federal-Aid Highway Act of 1956 or the Interstate Highway Act ushered in an era of unprecedented development and expansion, often at the expense of the same communities already caught in the historical crosshairs of existing discriminatory practices (Archer, 2020).

Environmental justice

In the context of policy, the U.S. Environmental Protection Agency (EPA) defines environmental justice as the fair treatment and meaningful involvement of all people, regardless of race, color, national origin, or income, in the development, implementation, and enforcement of environmental laws and policies (EPA, 2019).

Through a social lens, environmental justice may also be thought of as a means of examining both the outcomes and the related origins of inequality through which a framework of change can be cooperatively constructed and executed (Bullard and Johnson, 2000). For example, the Bullard and Johnson (2000) framing of environmental *in*justice - (1) unequal enforcement of environmental, civil rights and public health laws; (2) differential exposure of some populations to harmful chemicals, pesticides, and other toxins; (3) faulty assumptions in calculating, assessing, and managing risks; (4) discriminatory zoning and land use practices; and (5) exclusionary practices that prevent some individuals and groups from participation in decision making or limit the extent of their participation. This framing offers an alternative perspective through which to measure outcomes within the CHANS, rather than solely through the physical environment with weight given to the way these manifest and compound to disproportionately impact already vulnerable populations.

The University of Michigan's 2020 Environmental Justice Fact Sheet (UMich, 2020) describes environmental *in*justice in terms of, for example, greater exposure to pollution, loss of land, property or rights to resources, or limited access to environmental services and delineates five (5) categories in which marginalized communities may experience these negative impacts: (1) the built environment, (2) through limits of food access, (3) energy poverty, (4) materially, and (5) climate change impacts. What this report also notes is the inextricable link between environmental justice and sustainability. While Robert Bullard defined environmental justice as the principle that all people and communities are entitled to equal protection of environmental and public health laws, the environmental justice movement has broadened the definition of environmentalism to include all aspects of life, both physical and cultural environments, and emphasized the integration of justice across all activities. Included in this understanding of environmental justice is the disproportionate exposure of already vulnerable communities to poor air quality and its attendant effects. This scholarship considers this framework of exposure and justice within the Eastern Shore region of Maryland.

Case Study Area - The Georgetown Community of Salisbury, Maryland

The Georgetown neighborhood in Salisbury, Maryland, historically known as Lot #10, represents a significant chapter in the city's African American heritage. Situated around Humphrey's Lake prior to 1909, Georgetown emerged as a vibrant Black community with economic and cultural influence; it reflected a significant portion of Salisbury's African American community, with records from 1907 indicating that the community made up approximately 25% of the city's total population (Duyer, 2007).

The community boundary originated near the east prong of the Wicomico River and expanded following the collapse of the Division Street dam in 1909, which led to the drainage of the lake and further real estate development (APPENDIX E). This expansion fostered the growth of Black-owned businesses and institutions, reinforcing the neighborhood's role as a thriving enclave for African American residents; proximity to the railroad tracks further cemented its importance as a hub for Black economic and social life.

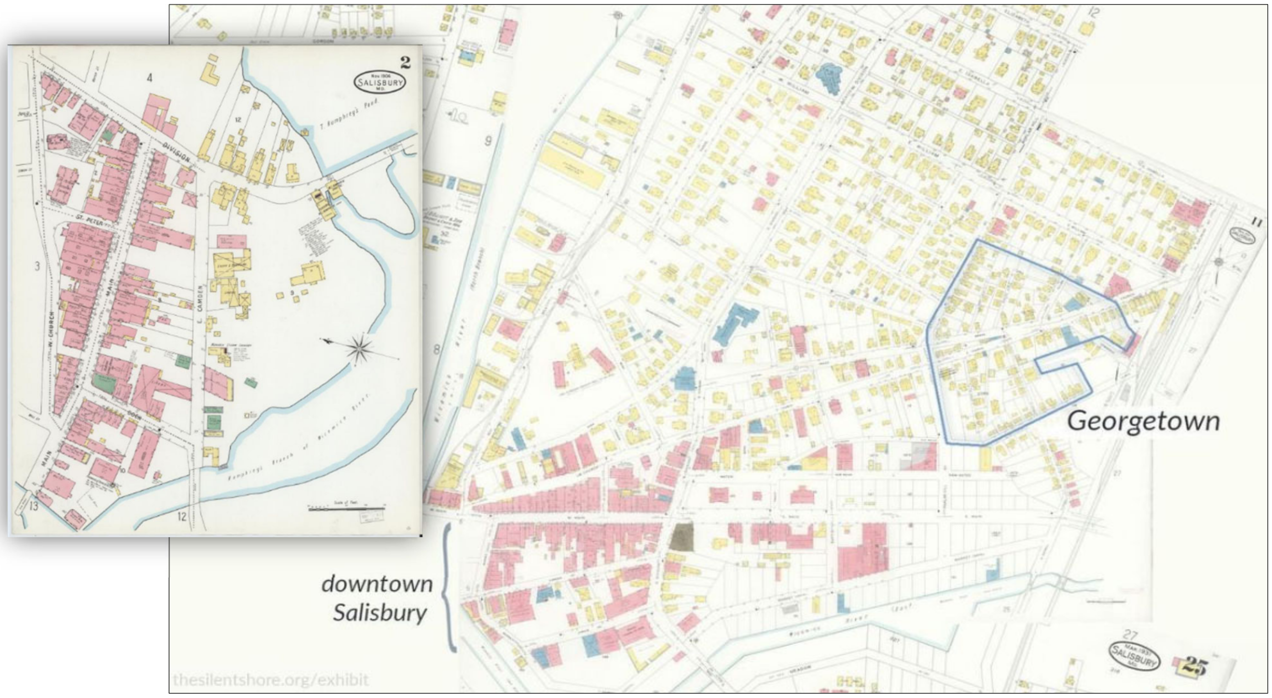

Figure 2a Georgetown section of Salisbury, Maryland (blue outline), Humphrey's Pond before the flood (cut out, upper left). Following the 1909 dam break, Thomas Humphrey sold former pond location to property developer, Salisbury Realty Company (Duyer, 2007, MAP: Stancil, 2022)

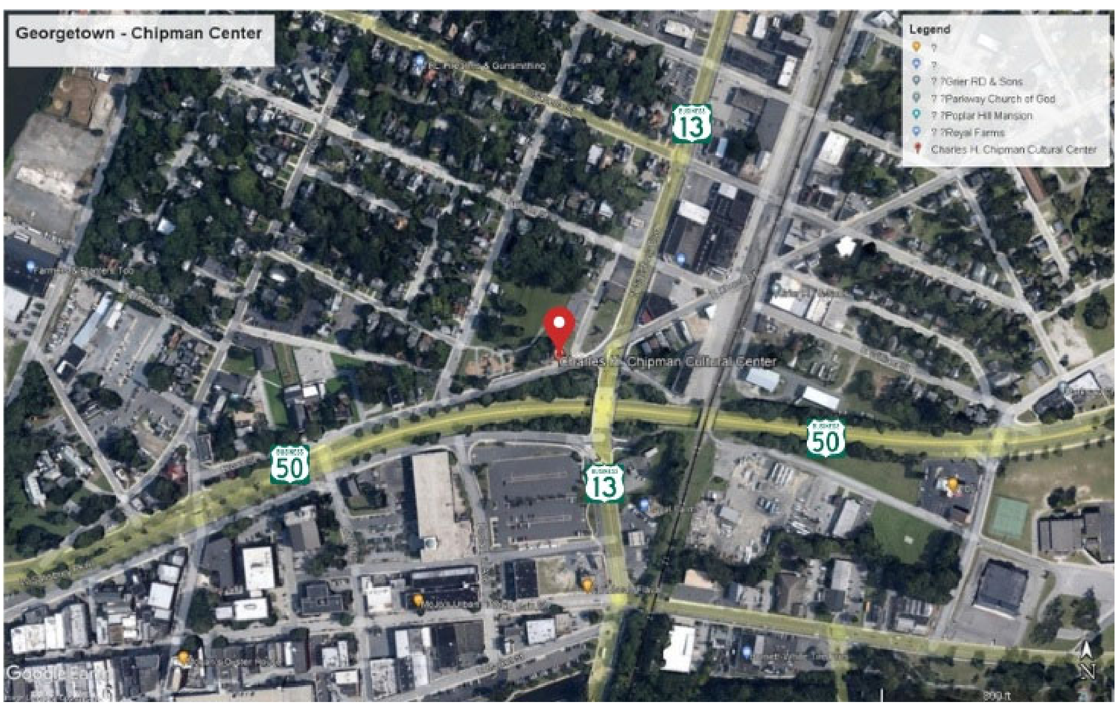

Figure 2b Georgetown map section, Google Earth 2025. US Routes 13 Business and 50 Business, Charles H. Chipman Cultural Center, location icon, former site of John Wesley A.M.E. Church (Stancil, 2022)

The Georgetown area was a walkable community that boasted a strong network of churches, businesses, and social organizations of historical and cultural significance to the region (Duyer, 2007). However, with the increase of major highway projects following the passage of the Federal Highway Act of 1956, the cohesion of this historic community would be increasingly disrupted.

Between 1948 and 1965, highway expansion projects like those of U.S. Routes 13 BUS and 50 BUS carved through Salisbury, forcibly relocating African American residents from the Cuba and Georgetown neighborhoods. In the process, residents would be dispossessed of their homes and businesses; over 889 graves were even exhumed to make way for the new highway (Bruder, 2010). Residents of these newly fragmented communities were not only displaced but were often exposed to increased environmental hazards such as air pollution from vehicle emissions. These actions were part of broader state and federal initiatives that privileged traffic efficiency and economic growth over the welfare and stability of minoritized populations.

What occurred in Georgetown was not an isolated example; similar patterns of disruption occurred in numerous cities including Camden, Milwaukee, Richmond, and Syracuse with immediate and devastating consequences for these historically Black neighborhoods. For example, In St. Paul, Minnesota, the Rondo community was dismantled to accommodate Interstate 94, while in Miami, Florida, I-95 tore through Overtown, a thriving cultural and economic hub for Black residents. These communities were often strategically targeted under the pretense of “slum clearance” or “urban renewal,” language that masked the displacement and dispossession of vulnerable populations in the name of progress (Mahajan, 2024; Archer, 2020).

Despite its physical erasure, Georgetown's legacy persists through historical preservation efforts, oral histories, and surviving landmarks like the Charles H. Chipman Cultural Center. The neighborhood's story highlights the resilience of Salisbury's African American community and underscores the importance of recognizing and preserving historical neighborhoods affected by systemic displacement. Georgetown's history serves as a crucial reminder of the broader social and environmental impacts of urban development on marginalized communities, and the increased traffic volume following installation of US Routes 13 and 50 present a unique case study for the effects of $PM_{2.5}$ vehicle emissions and chronic disease.

$PM_{2.5}$ Air Pollution

Particulate matter (PM), dynamic compound of atmospheric aerosols, has a wide variation of complex properties; they range in size from (PM) and $PM_{2.5}$ with the most problematic being the smaller forms, typically PM_{10} and $PM_{2.5}$. While PM_{10} are a nuisance with an aerodynamic diameter ≤ 10 μm, like airborne dust, eye irritants, or pollen and contribute to largely external irritations, $PM_{2.5}$ is a microscopic mixture with an aerodynamic diameter ≤ 2.5 μm; it has the ability to enter the lungs, bloodstream, and be absorbed into soft tissue and contribute to neurological dysfunction (Braithwaite, et al, n.d.). Among its contaminants are carbon dioxide (CO_2), lead (Pb), arsenic (As), copper (Cu), cadmium (Cd), nickel (Ni), barium (Ba), and organic nitric dioxide ($NO_{x)}$ and sulfur dioxide (SO_2); these aerosolized particulates pose a threat to internal organs, biological and neurological tissue formation, affecting gene expression, cell formation, and fetal development. Its small size penetrates respiratory cell tissue and enters the bloodstream compromising cell viability and organ function which can contribute to chronic diseases and cancer.

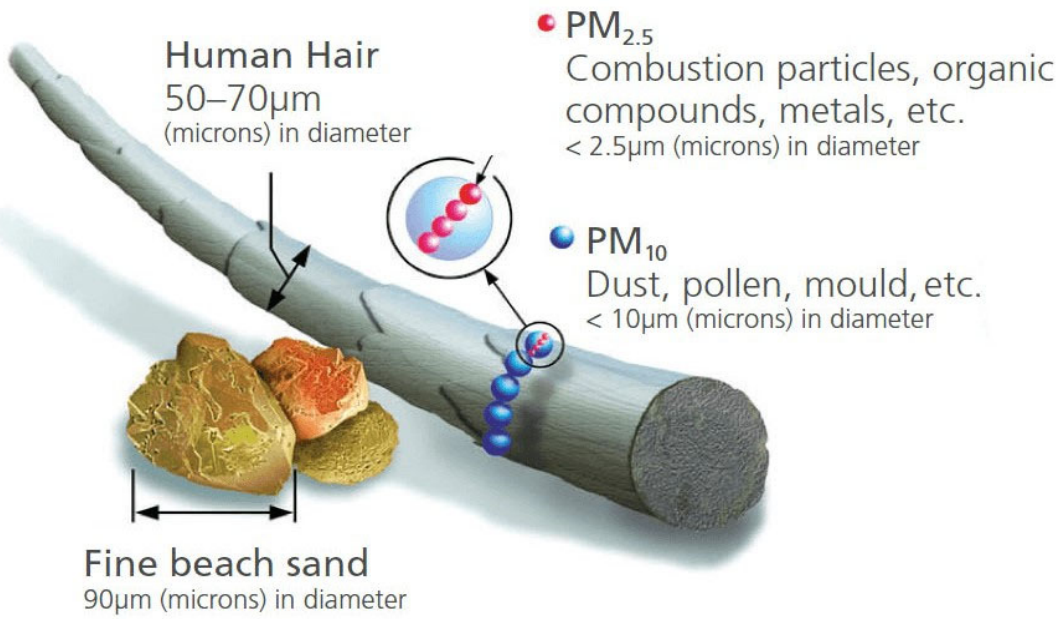

Figure 3 - Particulate Matter, EPA.gov

The concentration and distribution of $PM_{2.5}$ in the atmosphere are shaped by a complex interplay of geographical, meteorological, and anthropogenic factors. In urban and suburban areas, population density, land use, and impervious infrastructure often amplify or interfere with natural ecological processes, leading to distinct air quality challenges. The spatial layout of the built environment can trap pollutants and inhibit their dispersion, especially during atmospheric events such as temperature inversions that reduce vertical mixing. For example, core compounds comprising $PM_{2.5}$ like sulfates (SO_4^{2-}), nitrates (NO_{3}-), ammonia (NH_3), and sodium chloride (NaCl) are augmented in lower temperatures of the winter; and nitrogen oxides (NO_x) and sulfur dioxide (SO_2) have been found to increase in response to meteorological phenomena (Jandacka et al., 2018). These dynamics are particularly evident in regions where natural landscapes intersect with dense human settlement.

On the Lower Eastern Shore of Maryland—comprising Somerset, Wicomico, and Worcester counties—the region's coastal geography and land use create unique environmental conditions that influence $PM_{2.5}$ behavior. Characterized by a mosaic of flood plains, aerated agricultural fields, comingled with densely built urban centers, this region is vulnerable to seasonal and spatial variations in air pollution previously described. For example, coastal influences introduce warm, humid air masses and dynamic wind patterns, while seasonal weather phenomena such as extreme flooding and winter temperature inversions contribute to the persistence of airborne pollutants (Han et al, 2023). Low winter temperatures have been shown to enhance the concentration of $PM_{2.5}$ components such as sulfates, nitrates, ammonia, and sodium chloride, while nitrogen oxides (NOx) and sulfur dioxide (SO_2) levels also respond to regional meteorological changes (Jandacka et al., 2018). In addition, emissions from concentrated animal feeding operations (CAFOs), prevalent in the region's agricultural sector, further compound the area's pollutant burden (Bediako and Sauder, 2023).

Understanding seasonal and regional variability in $PM_{2.5}$ is essential not only for environmental monitoring but also for assessing public health outcomes. Fine particulate matter contributes to a range of chronic health conditions, including cardiometabolic diseases. Recent studies have employed satellite data, surface-level measurements, and atmospheric chemical models to assess population-level exposure to $PM_{2.5}$ and its link to health outcomes such as diabetes. Using metrics such as years lived with disability (YLD), years of life lost (YLL), and disability-adjusted life years (DALYs), researchers have quantified the burden of disease attributable to $PM_{2.5}$ concentrations exceeding theoretical minimum risk exposure levels

(TMREL). These data-driven approaches reveal that long-term exposure to $PM_{2.5}$ can substantially elevate the incidence and severity of chronic diseases across affected populations (Bowe et al., 2018).

Sources of $PM_{2.5}$

The sources of $PM_{2.5}$ have natural and anthropogenic components that are comprised of and distributed by chemical processes occurring in the environment. There are three (3) notable types and characteristics of PM associated with atmospheric sources of PM - (1) *nucleation* mode, (2) *accumulation* mode, and (3) *coarse* mode (Guo, et al, 2020).

The (1) *nucleation* mode consists of particles formed from gaseous precursors, typically with sizes less than 50 nm. Nucleation events frequently occur in areas with a high emission of biogenic volatile organic compounds, where particle formation can be influenced by the presence of existing particles and gaseous molecules. 'Nucleation' mode is the initiating process by which ions, atoms, and molecules form into crystalized solids to which additional particles may become attached. This mode is characterized by particles resulting from chemical reactions to ambient gases or combustive events/sources such as industrial or vehicular emissions or biomass burning. Nucleation mode particles are typically within the ultrafine particles (UFP) size range (<0.1 µm) and make only a nominal contribution to total PM mass and surface area; however, their small size better facilitates rapid uptake by gases and coagulation leading to accumulation mode (Hakala et al, 2022).

(2) *accumulation* mode with particles ranging between 0.1 µm and 1 µm. They result from fine particle emissions and dynamic processes like condensation and coagulation. The accumulation mode has a long atmospheric lifetime due to inefficient removal mechanisms, and

"*accumulation mode*" is characterized by an increase in PM mass and surface area to particles typically $_{2.5}$ µm, although some particles can exceed this size (Zhang et al, 2022). At this stage, the structure of the particle is stabilized and accumulates in this atmosphere as $PM_{2.5}$.

The final of the three is *coarse* mode, during which coarse particulate matter (>10 µm) results from processes such as sea spray formation and resuspension of dust, these particles are typically referred to as PM_{10} *coarse* mode in which particles have an aerodynamic diameter greater than 1 µm and are primarily emitted through mechanical processes. Coarse particulate matter typically contributes less to the total number of particles but can significantly affect visibility and health due to their larger size.

Another source of PM, emissions from motor vehicles have two main sources: the burning of fossil fuels, which releases emissions through tailpipe exhaust, and non-exhaust processes like the deterioration of vehicle components, road surfaces, and the stirring up of road dust. These processes collectively produce what is known as non-exhaust PM emissions. These emissions have a significant impact on urban areas, with PM concentrations often surpassing safe levels for human health. The toxicity of these particles is influenced by factors such as their weight, number, size, and chemical composition. There are also changes in the concentration of fine particles ($PM_{2.5}$), ultrafine particles (PM1), and coarse particles ($PM_{2.5–10}$) over time such as with annual variations, meteorological conditions, and anthropogenic activities that impact pollution levels. For instance, key constituents of PM like sulfates, nitrates, ammonia, sodium chloride, black carbon, mineral dust, and water have been shown to fluctuate during colder weather; in winter pollutants like SO_2, NO_x and aerosols are elevated due to increased emissions and meteorological factors like low temperatures and inversions, which hinder pollutant dispersion. Seasonal shifts in PM concentrations can also indicate diverse sources of air pollution within a given region.

Chronic Disease and $PM_{2.5}$

Fine particulate matter air pollution such as $PM_{2.5}$ poses a threat to internal organs and biological tissue and is associated with higher mortality rates for those with cardiovascular disease and diabetes (Bowe, et al, 2016). Studies like one by Hassanvand et al. (2006-2011) indicated a potential association between long-term exposure to PM_{10} and higher risk of developing T2DM. The cohort study by Li et al (2019) found that extended exposure to high $PM_{2.5}$ levels may elevate the risk of T2DM (Li et al, 2019). However, while many studies support this association, factors like population growth and aging also contribute to T2DM, and not all studies have consistently found increased risk.

According to the Global Burden of Disease (GBD) Study 2017, air pollution ranked as the eighth leading cause of death globally, contributing to 2.94 million deaths due to particulate matter pollution. T2DM, a metabolic disorder caused by genetic and environmental factors, leads to insufficient insulin secretion and impaired effects. Air pollution has been linked to factors like impaired glucose metabolism, insulin resistance (IR), and T2DM (Li et al, 2019).

The emissions of particulate matter (PM) from motor vehicles have two main sources: the burning of fossil fuels, which releases emissions through tailpipe exhaust, and non-exhaust processes like the deterioration of vehicle components, road surfaces, and the stirring up of road dust. These processes collectively produce what is known as non-exhaust PM emissions. These emissions have a significant impact on urban areas, with PM concentrations often surpassing safe levels for human health. The toxicity of these particles is influenced by factors such as their weight, number, size, and chemical composition (Ahmed, 2018).

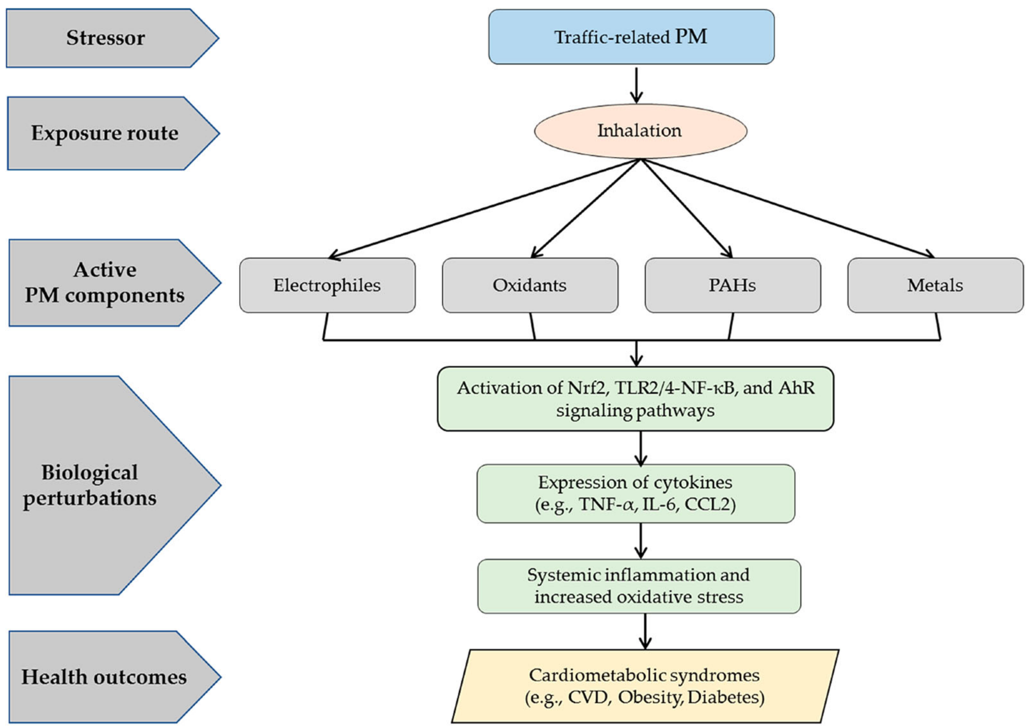

Figure 4 Traffic-related $PM_{2.5}$ chronic disease mechanisms, Ahmed et al (2018)

There are well-established links between specific elements in particulate matter (PM) and health issues. Aluminum and silicon have been connected to respiratory problems, while elements like iron, copper, zinc, and sulfur have been associated with adverse health effects, including cardiopulmonary oxidative stress, heart rate variability, and tissue damage.

Prevailing research also suggests a link between exposure to air pollutants and an increase in biomarkers related to type 2 diabetes mellitus (T2DM). The impact is influenced by the duration and concentration of air pollutants, and the chemical composition of pollutants (Li et al, 2019).

The GBD report indicates that approximately 7.5% of global deaths (11.1% in China) are linked to air pollution, with a 10 μg/m³ increase in $PM_{2.5}$ concentration leading to a 0.68% rise in all-cause mortality. Notably, even $PM_{2.5}$ exposure below current WHO standards still increases overall mortality rates and specific causes of death. Fine particulate matter correlates with adverse health effects like respiratory and cardiovascular diseases, and lung cancer. In 2019, 6.2 million deaths in the 30-70 age group were attributed to cardiovascular diseases (CVD). The American Heart Association asserts $PM_{2.5}$ as a controllable risk factor for cardiovascular events. Gender, age, and climate impact air pollution effects on health, with CVD linked to air pollution-related mortality.

Studies show that $PM_{2.5}$ triggers oxidative stress, nitric oxide production, and lipid peroxidation in cardiovascular endothelial cells (Qinglin et al, 2021). This interaction alters cell membrane fluidity and triggers plaque formation, causing atherosclerosis and cardiovascular impairment. Epigenetics, involving genetic changes in phenotype without DNA sequence alteration, plays a vital role in CVD occurrence and development. Recent research associates epigenetic regulation with cardiovascular risks post $PM_{2.5}$ exposure, including histone acetylation modification and interferon γ methylation. $PM_{2.5}$ enters the alveoli through the respiratory tract, penetrates the air-blood barrier and enters the bloodstream causing cardiovascular and respiratory diseases.

Exposure to PM air pollutants is also associated with an increase in biomarkers related to type 2 diabetes (T2DM) as available research suggests. One theory of mechanisms of $PM_{2.5}$ induced T2DM includes the release of inflammatory factors and cytokines due to exposure to air particles. This triggers inflammation, reduces insulin sensitivity, and hinders glucose uptake in peripheral tissues, thereby elevating T2DM risk. Oxidative stress due to reactive oxygen species (ROS) generated through exposure to $PM_{2.5}$ could also harm pancreatic β cells, contributing to T2DM. The impact is influenced by the duration and concentration of air pollutants, and the chemical composition of pollutants.

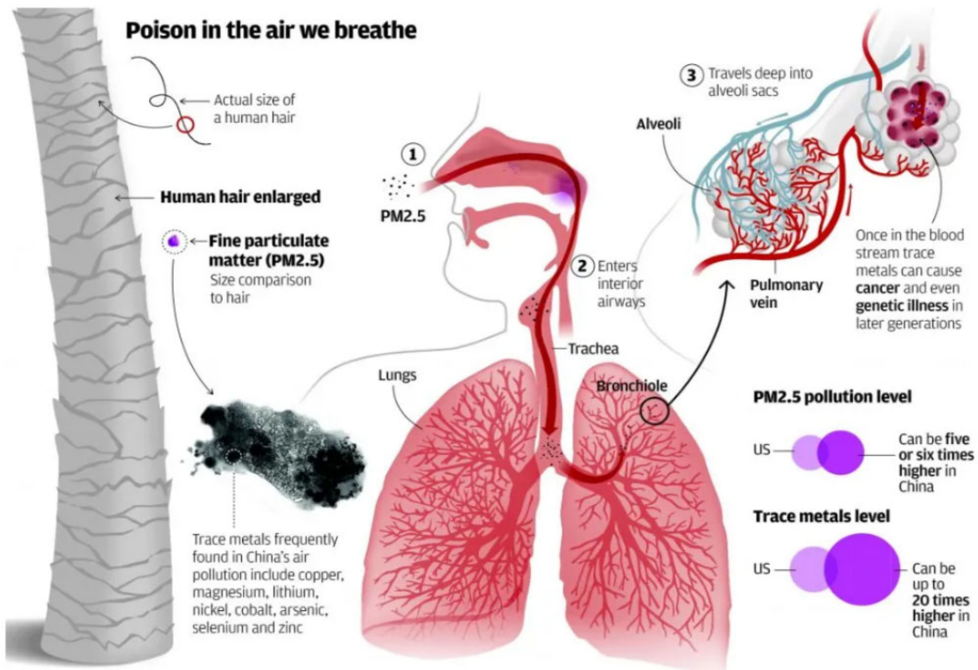

Figure 3 $PM_{2.5}$ Chronic Disease Mechanisms, Bobicic, et al. (2023)

A nonlinear exposure-response function was, however, established by Bowe et al. (2018) and showed a considerable T2DM risk increase with $PM_{2.5}$ concentrations beyond 2.4 μg/m3, with a moderate increase above 10 μg/m3. In contrast, researchers like Andersen found no significant association between air pollution and confirmed T2DM cases in a cohort study of Copenhagen or Arhus residents.

Air pollution, particularly fine particulate matter (PM) resulting from sources like fossil fuel combustion and biomass burning, has been increasingly associated with elevated BP (Zhao et al, 2022). Short-term PM exposure has been linked to rapid BP increases, and some studies suggest that chronic exposure in polluted regions might sustain this effect, potentially promoting hypertension development. Gaseous pollutants like ozone and nitrogen dioxide, while having less evidence, also appear to elevate BP (Brook et al, 2011).

Air pollution, a major global public health concern, affects a large population involuntarily and ranks as a leading cause of mortality. Even relatively low levels of PM found in cleaner countries elevate BP, with developing nations facing a disproportionately high burden due to higher pollution levels resulting from factors like rapid industrialization and poor indoor air quality.

Hypertension or high blood pressure (HBP) is influenced by various environmental factors such as ambient temperature, altitude, latitude, noise, and air pollutants like $PM_{2.5}$ (Bruno et al, 2017). These factors could potentially impact the prevalence and control of hypertension. While not established as modifiable risk factors like obesity, healthcare providers should recognize the significant role of the environment in altering BP and consider reducing exposures when relevant to prevent or control hypertension.

Using Global Burden of Disease (GBD) methodologies, this IER model synthesizes data on diabetes and $PM_{2.5}$ thus providing an integrated exposure–response (IER) function for evaluating the global diabetes risk associated with $PM_{2.5}$ exposure levels, as well as proxy exposures from passive and active smoking to bridge gaps in high-exposure data, often unavailable for $PM_{2.5}$ in developed nations. Smoking data was especially critical as it allowed for dose-response comparisons, with studies meeting strict inclusion criteria, including a minimum score of four on the (Bowe et al., 2018). The IER model adopts Bayesian hierarchical modeling to align with GBD methods and establish a theoretical minimum risk exposure level (TMREL) for $PM_{2.5}$, defined by the lowest observed risk levels between 2.4 $\mu g/m^3$ and 5.9 $\mu g/m^3$.

Overall, it is essential to consider environmental factors, especially air pollution and SHS, in understanding and managing chronic diseases like hypertension, diabetes and cardiovascular function.

Race and $PM_{2.5}$

There is well-established research documenting the racial disparities regarding $PM_{2.5}$ and PM_{10} pollution exposure, especially between Black people and Whites, that outweigh poverty-based disparities. For example, Mikati, et al (2018) build upon existing research that indicates a higher likelihood of non-Whites and individuals living in poverty residing near sites emitting $PM_{2.5}$. Their analysis reveals significant inequalities in PM-emitting facilities' impact on different demographic groups.

As with many vulnerable populations, many properties and agricultural lands are located adjacent to major roadways. U. S. Routes 50 and 13 span and transect Delaware, Maryland, and Virginia with a high volume of commuter and commercial vehicles traversing these interstate routes daily, exposing the soils of this rural-agricultural corridor of commercial crops of corn and soybean to fine particulate matter air pollution known as $PM_{2.5}$ (Ahmed, 2018). This scholarship considers these potential trends within the Eastern Shore region of Maryland.

The role of technology in environmental health management

The dearth of detailed minority health data was sharply exposed during the COVID-19 pandemic prompting key public health institutions including the National Academy of Sciences, the National Institutes of Health, and the National Science Foundation to activate multiple streams of research funding and opportunities that called for more collaborative, and interdisciplinary efforts to understand racial health disparities. Out of these calls to action, emerged frameworks like the *Exposome* with opportunities to expand multidisciplinary research, to reorganize evaluative priorities, and leverage existing and nascent technologies such as "Big Data" methodologies for improving environmental health analysis and outcomes.

An example of this new approach is the EXPOsOMICS research project; it employs Big Data methodologies characterized by the three Vs: volume, velocity, and variety searching for biological markers, or biomarkers, which are measurable indicators that reflect biological processes or exposures; researchers aim to find associations between biomarkers of exposure and biomarkers of disease, using databases like the Human Metabolome Database (Canali, 2016).

The EXPOsOMICS (2016) approach, rather than using traditional formal causal models, exposomics researchers employ statistical tools and causal knowledge to handle the complexity of their data and target systems; this includes variable selection models, univariate models, and multivariate approaches like Principal Components Analysis (PCA) to reduce data dimensionality and enhance computational efficiency. (Canali, 2016). Causal knowledge is indispensable for handling Big Data complexity and validating the results, however, opponents debate the effectiveness of these methodologies highlighting the complexity and need for additional advanced data analysis techniques.

Big Data research methodologies increasingly require the use of Artificial intelligence (AI) and as such AI has significantly impacted the field of healthcare, particularly in clinical research, where patient data plays a crucial role in medical analysis. In such cases, AI is used with patient data that is typically categorized into baseline traits such as age, gender, and disease history, and disease-specific data and may also include diagnostic imaging, gene expressions, physical examination results, and clinical symptoms. Additionally, medical outcomes such as survival times and tumor sizes are recorded. The data gleaned is often represented in a machine-actionable structure by using a data matrix, with *Xij* denoting patient traits, where "*i*" refers to the patient and "*j*" refers to a specific trait, while the outcomes of interest are represented by *Yi* (Jiang et al., 2017). Attention to data interoperability is an evolving requirement in technology-assisted environments and will need to advance and adapt as AI technology advances.

The study by Walker et al. (2022) demonstrates the application of a high-resolution exposomics and metabolomics approach to elucidate potential environmental and metabolic contributors to cholestatic liver diseases, specifically primary sclerosing cholangitis (PSC) and primary biliary cholangitis (PBC). Another exposome-wide association studies (EWAS) identified associations between PSC/PBC and environmental agents such as pesticides, additives, and persistent pollutants, implicating these factors in disease progression. Concurrently, metabolome-wide association studies (MWAS) revealed disruptions across various metabolic pathways, including amino acid, eicosanoid, lipid, and mitochondrial metabolism, indicating shared pathophysiological features between the two diseases. A key contribution of this research is the integration of EWAS and MWAS through network analysis, which highlights the interplay between environmental exposures and metabolic dysregulation, potentially uncovering mechanistic insights into PSC and PBC pathogenesis.

Tania et al. (2022) used a proteogenomics approach, by leveraging urine analysis to uncover novel protein biomarkers; researchers employed an analytical process called ‘shotgun proteomics’ in which specialized computational environments were used (Santos et al, 2022) to characterize the urinary proteome, identifying eighteen differentially abundant proteins. The integration of proteogenomics into exposomic research provides a richer interpretation of disease mechanisms by correlating environmental exposures with molecular alterations.

Linear regression-based statistical methods have also been applied in exposome research to elucidate the complex interactions between environmental exposures and health outcomes and address the multifactorial nature of environmental influences by considering multiple exposures simultaneously, thereby mitigating the risk of selective reporting and false discoveries (Agier et al., 2016). Methods applied include the environment-wide association study (EWAS), elastic net (ENET), sparse partial least squares regression, Graphical Unit Evolutionary Stochastic Search (GUESS), and the deletion/substitution/addition (DSA) algorithm.

Finally, the research project, "WormJam: A Consensus C. elegans Metabolic Reconstruction and Metabolomics Community and Workshop Series" highlights the collective effort to standardize and expand metabolic reconstructions for *Caenorhabditis elegans* to address the fragmented and incomplete nature of existing metabolic models by merging independent reconstructions and curating metabolomics data. A transparent nematode found in temperate soils, *C. elegans*, is widely used in biological and medical research due to its genetic accessibility and similarity to higher organisms, serves as an ideal exposomic model for studying metabolism's role in healthspan and lifespan (Hastings et al., 2017). Despite the organism's relevance, metabolic reconstructions have lagged, with gaps in pathway coverage and inconsistencies across models.

AI in Environmental Health

Artificial Intelligence (AI) is a branch of computer science dedicated to developing intelligent machines capable of mimicking human thought and behavior. AI systems are designed to learn from their environment and make data-driven decisions, enabling applications such as medical diagnosis, autonomous vehicles, and natural language processing. Additionally, AI can contribute to cost reductions in information systems, electrical systems, and customer service operations. The field of AI has a long history, dating back to ancient Greek speculation about intelligent machines, but its modern development began in 1956 at Dartmouth College, where scientists and mathematicians convened to explore the creation of human-like computers. Since then, AI has progressed significantly, with innovations in machine learning, natural language processing, and robotics, influencing sectors such as healthcare, finance, retail, and transportation.

There are multiple types of AI, including machine learning, deep learning, and natural language processing (NLP). Machine learning involves using algorithms to analyze data and make predictions, while deep learning, a subset of machine learning, employs neural networks to process vast amounts of information. NLP, on the other hand, enables machines to understand and generate human-like conversations (APPENDIX D). One prominent example of NLP technology is ChatGPT, an AI system developed by OpenAI. ChatGPT, based on the GPT-3 deep learning model, is trained on extensive conversational datasets, allowing it to generate contextually relevant and human-like responses in dialogue (Deng et al, 2022).

Several exemplary case studies illustrate the transformative impact of AI across various sectors. For example, in a study regarding project management, ML algorithms were employed to predict cost overruns in construction projects by analyzing a dataset containing numerous independent variables related to project specifics (Uddin et al., 2022). Additionally, another case study applied AI to identify early life environmental exposures associated with health outcomes in children, revealing critical insights into risk factors for chronic diseases (Guimbaud et al, 2024).

With extensive applications in numerous fields, including healthcare, finance, marketing, and environmental science, AI applications are increasingly used in environmental health research to analyze complex datasets related to environmental exposures and their health impacts. For instance, AI models are utilized to predict environmental toxicities, assess human exposure to pollutants, and identify unknown hazards effectively (Olawade et al., 2024). Unsupervised learning methods, such as clustering and principal component analysis (PCA), are particularly useful in organizing and reducing complex data sets. Clustering algorithms, including k-means clustering, hierarchical clustering, and Gaussian mixture clustering, group patients with similar characteristics without considering the outcomes. PCA is commonly used in genomics to reduce the dimensionality of high-dimensional data, retaining critical information while simplifying the analysis (Jiang et al., 2017).

Randomized controlled trials (RCTs) like those implemented in healthcare environments, are used for better incorporating and understanding the nonmedical factors influencing health outcomes (Kessler, 2011). Although well-suited for specific purposes, pharmacology trials have limitations when addressing broader healthcare issues due to their narrow focus. The authors propose a ten-year moratorium on such trials and advocate pragmatic, transparent, contextual, and multilevel designs citing that designs should incorporate replication, rapid learning systems, mixed methods, simulations, and economic analyses to yield actionable and generalizable findings for real-world implementation. Dominance of RCT designs and systematic reviews have excelled at initiating research; however, they have been less successful with generating replicable or applicable results. While these studies have been largely applied machine learning (ML) to existing datasets, other applications have combined AI technologies including large language models (LLMs), natural language processing (NLP) and optical character recognition (OCR) for typewritten and handwriting extraction tasks.

AI and the Evolution of Large Language Models (LLMs)

Language Models (LLMs) are artificial intelligence systems that can process and generate text with coherent communication and generalize to multiple tasks (Naveed, et al, 2023). Historically, LLMs emerged from natural language processing (NLP), statistical then neural language modeling, and finally evolved into pre-trained language models (PLMs). While conventional language modeling (LM) trains task-specific models in supervised settings, PLMs are trained in a self-supervised setting on a large corpus of text with the aim of learning a generic representation that is shareable among various NLP tasks. By significantly increasing model parameters (tens to hundreds of billions) and training dataset (many GBs and TBs), PLMs improved and surpassed traditional language modeling (LM) which led to the transitioning of PLMs to LLMs.

The early work on LLMs employed transfer learning until GPT-3 showed LLMs are zero-shot transferable to downstream tasks without fine-tuning. LLMs accurately respond to task queries when prompted with task descriptions and examples. However, pre-trained LLMs fail to follow user intent and perform worse in zero-shot settings than in few-shot. Fine-tuning them with task instructions data and aligning with human preferences enhances generalization to unseen tasks, improving zero-shot performance significantly and reducing misaligned behavior.

Before 2017, natural language processing (NLP) models relied on supervised learning for specific tasks, limiting their versatility. The introduction of the Transformer architecture in 2017 revolutionized NLP, leading to the development of Bidirectional Encoder Representations from Transformers (BERT) and Generative Pretrained Transformer (GPT) in 2018. These models leveraged a semi-supervised approach, combining unsupervised pre-training with supervised fine-tuning, allowing them to generalize across various tasks. GPT models have since undergone rapid advancements, with GPT-3 reaching 175 billion parameters—100 times larger than its predecessor, GPT-2. However, despite their impressive text generation capabilities, LLMs like GPT-3 exhibit unintended behaviors, including fabricating information, reinforcing biases, and occasionally failing to follow user instructions. This misalignment arises because LLMs primarily predict the next text element based on extensive internet data, replicating existing biases and stereotypes.

Addressing the alignment problem remains a key area of research. OpenAI has implemented a moderation system to filter undesired content, such as hate speech, violence, and other controversial topics. A more significant innovation came with the development of InstructGPT, a model trained using reinforcement learning from human feedback (RLHF). Unlike GPT-3, which passively predicts text, InstructGPT, with its 1.3 billion parameters, was fine-tuned to follow user instructions in a safe and helpful manner.

The role of GeLU activation function is to improve ChatGPT's complex pattern recognition capacity; this more nuanced form of learning allows for small negative values to be smoothed and suppressed rather than 'zeroed out' helping with gradient flow and final character output.

With the rapid model convergence during training, key tasks like NLP perform with greater efficacy.

An additional component of ChatGPT includes its activation function. All versions of GPT employ the Gaussian Error Linear Unit or GeLU as its activation function as it mitigates the vanishing gradient problem and facilitates the generation of smoother gradients throughout the training process (M. A. K. Raiaan et al). Models like BERT, OPT, and T5 use ReLU as the activation function. Below is shown the non-linear activation function of GeLU (equation 1) and an approximation of the integral-based definition of the same (equation 2).

$$\mathrm{G}eLU(x) = 0.5x \cdot (\tanh\left(\sqrt{\frac{2}{\pi}}\ (x + 0.44715\ x^3)\right) + 1) \qquad \text{(equation 1)}$$

whereas the factor **0.5x** ensures that the function smoothly transitions between negative and positive values, the **tanh** helps approximate the Gaussian cumulative distribution function (CDF) or the probability of distribution normality and the cubic term adds to the efficiency approximating the integral form of GeLU.

$$\mathrm{G}eLU(x) = xP(X \leq x) \qquad \text{(equation 2)}$$

where *P(X*≤x)P(X \leq x)P(X≤x) is the probability that a Gaussian random variable is less than x.

Table 1 - Evolution of Large language models evolution toward ChatGPT

LLM	Year	Advantages	Disadvantages
GPT	2018	first model using semi-supervised training	trained on the BooksCorpus (800 M words)-need supervised fine-tuning to perform a specific task
BERT	2018	trained on BooksCorpus (800 M words) + Wikipedia (2,500 M words)-bidirectional architecture-unified architecture across different tasks	need supervised fine-tuning to perform a specific task
GPT-2	2018	trained on WebText (40GB of text) -learn to perform tasks directly without the need for supervised fine-tuning (task-agnostic)	zero-shot performance (without fine-tuning) still far from usable.
GPT-3	2020	trained on CommonCrawl (570 GB of text) + WebText + Wikipedia+ Books1-2 - trained using zero-shot, one-shot (one example of the task), and few-shot (10–100 examples of the task) settings	In text synthesis sometimes semantic repetition and loss of coherence over sufficiently long passages; retains the biases of the data it has been trained on.
Instruct-GPT	2022	aligned to act in accordance with the user's intention using reinforcement learning from human feedback	aligning to demonstrations and preferences provided by training labelers, not to human values. -follow the user's instruction, even if that could lead to harm in the real world.

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AI Handwritten Text and Optical Character Recognition (OCR) Function

The aim of this research is to use AI tools to extract data such as demographic, mortality, and other data from digitized images of certificates of death to examine the prevalence of deaths potentially related to $PM_{2.5}$ air pollution. These document images contain a complex array of typewritten and cursive handwritten text which will require a form of text-line extraction (TLE). Although has AI multiple functions for text analysis such as translation, keyword extraction, and optical character recognition (OCR) (APPENDIX B), TLE models can face challenges complex problems like character overlap, multi-skewed lines, and inconsistent inter-line spacing as with the death certificates being used in this study (APPENDIX A).

To address issues with text complexity, Kundu et al (2020) introduced a novel approach by using Generative Adversarial Networks (GANs) for TLE as an image-to-image translation task. Utilizing a U-Net generator and PatchGAN discriminator, they optimize their model with a combination of GAN loss, L1 loss, and L2 loss. Their method, evaluated on HITMW and ICDAR 2013 datasets, surpasses several state-of-the-art techniques, demonstrating their efficacy in overcoming TLE complexities (Kundu et al., 2020).

OCR and Neural Networks

While OCR models are proficient at converting standard text, they struggle with complex mathematical expressions, which often involve nested structures, symbols, and hierarchies requiring high-level semantic understanding to ensure accurate conversion. Tran et al. propose a Convolutional Recurrent Neural Network (CRNN) architecture for offline handwritten text recognition. Combining convolutional layers for feature extraction, Bidirectional LSTM (Bi-

LSTM) for sequence learning, and Connectionist Temporal Classification (CTC) for labeling, CRNN eliminates manual segmentation and enables global training. This approach is adaptable across languages and computationally efficient (Tran et al., 2019). Additionally, Pashine et al. compare SVM, MLP, and CNN for handwritten digit recognition using the MNIST dataset. While SVM employs hyperplane-based classification, MLP relies on feedforward layers, and CNN utilizes convolutional layers for feature extraction. Each model's performance is evaluated on accuracy and computational efficiency (Pashine et al., 2021).

In the Orji et al (2022) article “Analyzing the Limitations of Current OCR Models,” the authors examine challenges facing optical character recognition (OCR) models, particularly regarding the digitization of mathematical expressions into LaTeX.

The integration of AI into handwriting recognition emphasizes its ability to mimic human cognitive processes, exploring and showcasing how machine learning (ML) and artificial neural networks (ANNs), showcasing how these systems learn from datasets for tasks like character and digit recognition. For example, artificial neural networks (ANNs)are modeled after biological neural networks; these algorithms process inputs and adjust connections during training to generate outputs (Aqab & Tariq, 2020).

Another tool used for handwriting recognition is the application of Hidden Markov Models (HMM) combined with NNs. HMM employs probabilistic modeling to predict hidden sequences from observed variables while NNs leverage deep learning to recognize handwriting from images. HMM's statistical framework makes it particularly effective for handwriting recognition, as it can manage the variability and uncertainty inherent in handwritten data (Aqab & Tariq, 2020).

A crucial step in any machine learning-based handwriting recognition process is document digitization (Singh and Yadav, 2021). This process involves converting physical documents into digital images, followed by preprocessing steps like binarization, noise removal, skew correction, and skeletalization - steps that improve the clarity of the dataset and enhance recognition accuracy. Additionally, segmentation and feature selection further refine the data, while training, validation, and testing phases ensure the robustness and generalizability of the recognition model.

Special considerations for AI in Environmental Health

Data Quality

The use of Big Data methodologies like AI must include considerations for data quality – its origins, structures, and usability. The Fourches et al. (2015) study highlighted some of the risks associated with historical datasets (e.g., issues of incompleteness, inaccuracy, and format, incompatibility, etc.). and the need for tools and frameworks to enhance data reliability and reproducibility. Such a process is essential to AI’s role in public health, ensuring that models trained on these datasets yield valid and applicable results, fostering progress in chemogenomics and related fields. Additionally, AI's role in areas like molecular drug discovery is challenged by data limitations and reproducibility issues, underscoring the need for meticulous data management and regulation. Standardization in data generation and reporting is thus a necessary precursor to leveraging AI's full potential in this sector.

The management of data quality is also explored to understand its role as an interface between data governance and data literacy (Koltay, 2016). Data governance is defined as the exercise of decision-making and authority, involving a system of decision rights and accountabilities based on agreed-upon models that dictate who can take what actions, when, and under what circumstances, using specific methods.

The risk of corrupting research results due to incomplete, inaccurate, imprecise, incompatible, and irreproducible historical data, known as the 'five I's', highlights the importance of thorough data curation as a primary step in data analysis. To address data quality concerns, a chemical and biological data curation workflow is proposed (Fourches, et al, 2015). This workflow utilizes cheminformatics methods to identify and correct errors in large chemogenomics datasets. The workflow involves steps such as chemical data curation, duplicate analysis, assessment of experimental variability, exclusion of unreliable data sources, detection of activity 'cliffs', and calculation of a dataset modelability index. Consensus QSAR modeling is used for error identification and correction. The workflow aims to eliminate nonstandardized and duplicated structures, unreliable data, and structural outliers, enhancing data quality and reproducibility. It is recommended that experimental and computational scientists establish standards and best practices for data generation, reporting, and curation to improve data reliability and accelerate progress in chemogenomics research.

Data Management

In data-rich research environments, the concept of being ‘machine-actionable’ or machine interoperability is essential to the accessibility and utility of datasets. This term refers to two main aspects: first, the contextual metadata surrounding a digital object (i.e., what it is), and second, the digital object itself (i.e., how to process or integrate it) (Tedersoo, et al, 2021). A ‘machine-actionable’ state is a continuum where a digital object provides increasingly detailed information to an autonomous computational data explorer, like ChatGPT 4.0 or the recently emerged, DeepSeek. *Machine-actionable* explorations might include (1) identifying a type of object (its structure and intent), (2) determining its usefulness within the context of the current task by examining metadata and data elements, (3) assessing its usability based on license, consent, or other accessibility constraints, and (4) decision-making (i.e., what actions to pursue based on task identification). It is the scope, scale, and speed required by contemporary scientific data that is particularly relevant to the use of AI in research, healthcare, and environmental management disciplines, among others. The ability to employ machine-actionable technologies in data rich environments helps to extend beyond the limitations of human ability and the addition of metadata helps to provide context for AI algorithms to make better decisions about rank and relevance of one variable or source of data over another. - a process that necessitates a data standardization framework.

The F.A.I.R. data standards, an acronym for Findable, Accessible, Interoperable, and Reusable, are essential principles designed to enhance the usability and sharing of data across various domains, particularly in the context of scientific research (Wilkinson, et al, 2016). These standards facilitate the management and dissemination of data by ensuring that information is not only easily locatable but can also be effectively understood and utilized by both humans and machines which is critical for data mining, integration, and automated analysis—improves when data are structured according to FAIR guidelines. This interoperability is essential for harmonizing multidisciplinary research communication and advancing scientific discovery in data-intensive fields.

These principles emerged from collaborative efforts by diverse stakeholder groups, aiming to improve the usability of research data by humans and machines initially as a draft on Force11;

these principles continue to evolve, reflecting ongoing community discussions and implementations.

FAIR principles are domain-independent, making them applicable across various scientific fields and consist of high-level guidelines rather than prescriptive technical solutions, ensuring flexibility for diverse data types and publishing environments. This modular approach allows data producers to implement the principles incrementally, aligning with the evolving requirements of their domains.

By adhering to FAIR principles, researchers enhance the visibility and longevity of their data, facilitating collaboration and reproducibility. Machine interoperability —

The integration of artificial intelligence (AI) and machine learning (ML) technologies with F.A.I.R. data practices enhance data analysis capabilities across various fields. By ensuring that datasets are structured according to F.A.I.R. principles, AI tools can learn more effectively from diverse data sources, leading to improved predictive models and decision-making tools. This synergy between F.A.I.R. principles and advanced analytics is poised to drive innovations in areas such as healthcare, environmental monitoring, and public policy.

One such group facilitating these technology-enhanced approaches and data harmonization is the Environmental Health Language Collaborative (EHLC), established by the National Institutes of Health Environmental Health Service (NIEHS); this collective emerged from the recognition that, despite the existence of numerous language and data standardization efforts, the environmental health sciences community required a specialized platform to address its unique challenges (NIEHS, 2014). Two pivotal workshops, conducted in 2014 and 2019, laid the foundation by identifying critical needs and proposing actionable steps to improve data harmonization in the field. The EHLC was subsequently formed to provide a sustainable framework for ongoing development and coordination. This vital interdisciplinary initiative works to unify and standardize the language used in environmental health sciences (EHS). The overarching vision of the EHLC is to enhance the value of EHS data for knowledge discovery and translational research, promoting the consistent formatting of EHS data to enhance data discovery, interoperability, and reuse, ensuring that environmental health professionals can effectively communicate and share information.

Data Literacy

Data literacy, which is closely related to research data management (RDM), is a crucial component of academic, research and emerging data-related services. From frameworks around RDM, the concept of data literacy education continues to develop including discussions about their relationship with data governance, and data quality.

Data Ethics

Despite AI's promising applications, ethical considerations remain central to discussions about its role in environmental health. Yu, Beam, and Kohane (2018) highlighted inherent risks with AI's opacity, particularly when using complex algorithms like those in deep learning. Transparency challenges, especially regarding deep image analysis, remain topics of considerable debate. Concerns over patient data privacy and the risk of stigmatization of vulnerable populations are at the forefront of this complex ethical discourse as well as issues surrounding patient data ownership - individual ownership may improve patient engagement, however, a communal ownership model might better serve the goals of personalized medicine.

Western healthcare systems' shift towards individual patient responsibility has led to discussions about the ethical aspects of continuous monitoring through medical devices and their potential pitfalls. Although this monitoring could inadvertently stigmatize chronically ill or disadvantaged individuals and penalize those unable to meet new health standards, minimal attention has been given to these nuanced concerns in health policy debates.

At this writing, AI-based deep learning tools and their concomitant risks are unresolved; unregulated training sets, unsupervised learning constraints, data set size and variability, and patient confidentiality exposures will require a focus on developing more secure human-computer interfaces, while innovating solutions to mitigate technological limitations.

CHAPTER 2: METHODOLOGY

This study uses an exploratory mixed-methods approach that includes integrating AI, systematic archival research, and manual data extraction to investigate the impact of traffic-related $PM_{2.5}$ air pollution on chronic disease prevalence in the Georgetown community of Salisbury, Maryland. The research focuses on the periods between1911 to 1922.

Case Study Area

Background

Case Study Area: The Georgetown Community of Salisbury, Maryland

Subjects for this research are decedents from the case study area of Georgetown in Salisbury, Maryland in Wicomico County; The Maryland State Archives provided digitized certificates of death (CODs) for both periods. Records were sourced from Maryland Counties Series SE43 (1911-1922) using the onsite archival search room database in Annapolis, Maryland. To ensure a manageable dataset while maintaining statistical relevance, a systematic sampling approach was applied, selecting records from January, April, July, and October for each year within the target periods. In total, of the 1680 CODs from the earlier period and the 1173 CODs from the latter, a sample from 1911 to 1914 period was cleaned then analyzed.

The case study area was selected because of its significance as a multiracial but predominantly Black community, for its trackable proximity to U S Routes 13 and 50 and increasing traffic volume over time, and the prevalence of documented mortalities related to chronic diseases known to be caused or complicated by traffic-related air pollution. Because the boundaries of sample study area are more theoretical than geographical and geocoding addresses to decedents would re-identify anonymized subjects, assumptions were made about those who likely resided within the historical community of Georgetown.

Specific characteristics (e.g., age, sex, race, ethnic origin, religion, social or economic qualifications, cause of death) are considered relevant and were included in the data compiled albeit in an aggregated and/or de-identified form. All decedents, regardless of age, sex, race, ethnic origin, religion, social or economic qualifications, are being selected on the basis of having resided and expired in the case study region.

Manual Data Entry

For data analysis, decedent data for the period from 1911 to 1914 (n = 561) was manually copied from the death certificates and entered in an Excel spreadsheet manually. For each death certificate image collected, an entry was transcribed into a detailed 'codebook' selecting the following variables: death district number, date of birth, date of death, age at death, zip code, race, gender, marital status, occupation, and primary and secondary (if applicable) cause(s) of death. Included with each entry were detailed metadata, coding conventions, and medical billing classifications adapted from the International Statistical Classification of Diseases and Related Health Problems 10th Revision (ICD-10):version 2019 (ICD-10, 2019), North American Industry Classification System (NAICS) occupation codes (NAICS, n.d.) and researcher notes for clarification (e.g., modern translation of outdated terms, etc.).

Table 2 – Chronic Disease Classification, ICD-10 (2019)

Root node	Parent nodes	ICD Code	Child nodes
$PM_{2.5}$ related	Asthma	J45.9	Bronchial asthma
	Cancer	C43.52	Cancer
		C50	Cancer, breast
		C55	Cancer, uterus
		C80.9	Malignant neoplasm
	Cardiovascular	I20	Angina pectoris
		I38	Endocarditis
		I05.0	Mitral stenosis
		I46	Cardiac arrest, cardioplegia
		I50	Congestive heart failure
		I63	Cerebral infarction
		I63.3	Apoplexy
		I70	Atherosclerosis, chronic
	Diabetes	E14	Unspecified diabetes
			Diabetes mellitus
Not PM2.5-related			

In the codebook, the coding conventions were used to address incomplete or ambiguous information and to standardize the data for analysis. For example, on death certificates where the attendant did not provide the required metrics, entries are marked "UK" (unknown) or "NR" (not recorded), whereas "NA" (does not apply) was used for details deemed immaterial to research objectives. When the month and/or day of birth was missing, it was entered as "00/00/year," indicating an unknown or unrecorded birthdate; the decedent's approximate year of birth was calculated by subtracting the estimated age at death from the year of death.

Salisbury zip codes were assigned based on the assumption of historical segregation practices. The Georgetown Community which had a focal point of two main zip codes – 21802 and 21803 and the convention applied was that White decedents resided in 21803, located on the periphery of what was considered Georgetown, and Black decedents resided in 21802, located within the boundary and a standard error margin was estimated in the statistical analysis to account for deviations.

Occupation was recorded as "NA" for individuals below working age (.e.g, infants, school-aged children), retirees, and women not employed outside the home unless explicitly noted as "housekeeper" by the attendant; this assumption reflected traditional roles such as "housewife," "child," or "student" for the given time period.

For cause of death, blank entries for secondary causes were interpreted as the absence of contributory factors. Pathology duration was standardized using a unit metric of *years*, where durations of one year or less were recorded as "1 year," including shorter durations, such as "< 1 week" or "< 1 month,". These durations, recorded for the primary cause of death only, were aggregated into ranges for analytical purposes. The zip code field was also cross-referenced with the decedent's potential residence in historically significant areas, such as Georgetown. The attendant, identified as the individual completing the death certificate, may or may not have been the attending physician. These conventions ensured consistent data interpretation while aligning with the research objectives.

From the total of 1,680 certificates of death that were sourced from the Marland State Archives, the manually entered dataset comprised 561 records from the subsetted period of 1911-1914 that were entered into an Excel spreadsheet manually as an alternative to relying solely on AI to extract decedent details. From the 561 observations were extracted records containing causes of death the research has linked to $PM_{2.5}$-related chronic diseases (i.e., heart disease, hypertension, and diabetes, asthma, etc.) and excluded records in which key data was incomplete. The sample size was further reduced during the data cleaning process (i.e., de-identification and aggregation, removal of NA values, etc.) The final pre-highway sample size was n = 138 records.

Exploration of ChatGPT AI Tool for Data Extraction

This study explores the capability of ChatGPT 4.0 as a viable tool for handwritten text recognition and extraction to create novel datasets. Other available alternatives included deeper learning models and cloud API integrated OCR engines such as Google Vision, AWS Textract, and Azure OCR that are recognized for yielding greater accuracy and flexibility for printed material; however, with consideration given to the broader implications of data privacy and governance, these were not used.

Image preprocessing

For image preprocessing and statistical analysis, R programming language was used, leveraging several key packages including *tidyverse* for data wrangling and visualization, *tesseract* for OCR/text mining, and *dplyr* for advanced data manipulation; *ggplot* was used for data visualization. This comprehensive approach was intended to facilitate efficient analysis of the extracted data. COD images enhanced to improve image quality and facilitate higher accuracy in OCR outputs. Image files were resized to 3000x3000 pixels and underwent noise reduction using Gaussian filters. Additional steps included contrast enhancement, sharpening, and normalization. Processed images were saved in JPEG format for further OCR analysis.

Following image pre-processing, OCR was conducted using the *Tesseract* engine configured for English text recognition. Key handwritten and typewritten fields extracted from each COD included: date of birth, date of death, age, village or city of residence, race or color, sex, marital status, occupation, cause of death, contributory cause of death (if recorded by the attendant), and duration of illness.

ChatGPT 4.0

The AI tool used for this study was OpenAI's ChatGPT 4.0, selected for its extensive training horizon, accessibility, and proficiency with Optical Character Recognition (OCR) and handwriting interpretation tasks. A form of AI, ChatGPT 4.0, is a generative AI model based on GPT (i.e., generative pre-trained transformer) technology, has found applications across academia, medicine, Industry 4.0, education, and other fields (Alser and Waisberg, 2023). Advancements in GPT-based models highlight improvements in conversational capabilities, contextual understanding, and nuanced query handling (APPENDIX C) . These developments are pivotal for enhancing natural language processing tasks and widening the applicability of AI across sectors (Kalyan, 2024).

To evaluate the capabilities of ChatGPT 4.0 in processing historical records, a structured analysis was conducted using test datasets from multiple time periods. Initial tests utilized data from 1900, focusing on the extraction of critical demographic and mortality details such as the decedent's age, cause of death, occupation, and date of death. Three (3) separate extraction attempts were conducted, each refining the prompt engineering to optimize accuracy and consistency of data capture across different certificate formats with instructions that the extracted data be systematically organized into spreadsheets with demographic headers selected from the CODs based on the variables instructed in the provided prompt.

Building on the findings from the 1900 dataset, the extract task was expanded to encompass death certificates selected from 1911 and 1922. COD images were uploaded, and instruction prompts were refined to improve data extraction and enhance output accuracy.

To streamline data collection for large-scale analysis, (e.g., batch OCR/text extraction), a batch processing approach was adopted. Links to multiple death certificate images were provided, enabling automated extraction. The resulting data was collated into standardized spreadsheet formats, ensuring uniformity and compatibility across datasets.

CHAPTER 3: RESULTS AND DISCUSSION

The aim of this research was to use AI to investigate the relationship between race, traffic related $PM_{2.5}$, and its health effects in the case study community located within a 10-mile radius of major roadways. The results of this preliminary analysis indicate that AI, in this case, ChatGPT, has evolving potential to capture *Exposomic* contributors to chronic disease and to generate novel research datasets.

ChatGPT 4.0 Output

The first test document uploaded for ChatGPT 4.0 was in PDF form and included all records; I received a processing error message, and there was no output generated. In the second attempt, a pre-processed image of eight (8) certificates of death was provided in one PNG image document along with the instruction prompt. This attempt yielded better results. The third attempt was managed by providing ChatGPT 4.0 with a folder containing images and allowing for pre-trained refined extraction. In this iteration, the OCR extraction function auto corrected for noise and refocused on the handwritten details specified in the prompt. The results were varied and provided some insight into the navigational challenges AI experiences with character recognition and image quality.

ChatGPT's OCR capabilities for recognizing and extracting text from human cursive handwriting rely on deep learning, preprocessing and image enhancement, and contextual analysis and language models. It employs deep learning models, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), to identify and interpret patterns in handwritten text (Rajasekar et al, 2022). Cursive writing's variability in letter formation, spacing, and connectedness necessitates advanced pattern-matching techniques (Gonwirat et al, 2020) Certain letter combinations, such as *rn* vs. *m*, or *cl* vs. *d*, can appear almost identical when handwritten. CNNs are used for feature extraction, while RNNs are useful for sequence labeling and prediction tasks, such as handwriting recognition (Sak et al, 2014).

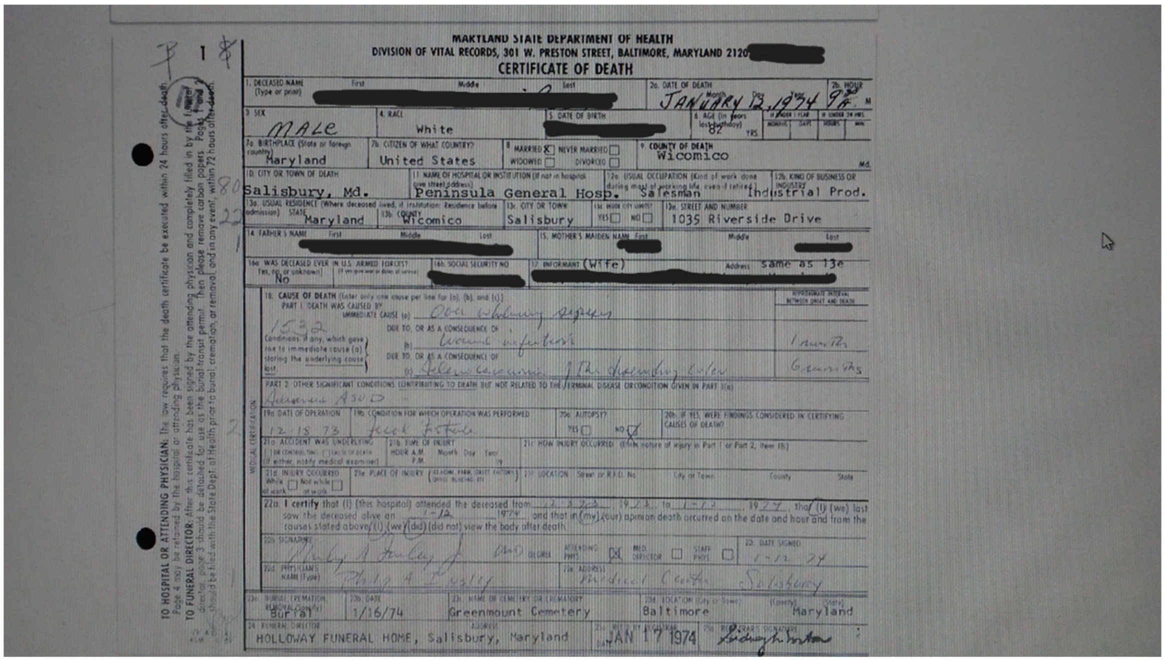
As mentioned, there are several key limitations of OCR models. Firstly, OCR technology struggles with handling complex equations that contain elements like variables, Greek symbols, integrals, and summations, all of which pose a challenge to achieving accurate representation in LaTeX format. Compounded by visual similarities between certain symbols (e.g., “0” vs. “o” or “x” vs. “×”), this similarity further complicates symbol recognition, requiring models to integrate contextual understanding to distinguish such characters accurately.

Unlike printed fonts, human cursive handwriting is highly variable, making it difficult to create a one-size-fits-all model (Shakib et al, 2023). Connected letters (ligatures) in cursive handwriting can obscure individual character. For example, in the word "flourish," the f, l, and o may be written as a continuous stroke, making it difficult to determine where one letter ends and the next begins. Letters like *g, y, f, j,* and *z* often have elongated loops that extend above or below the main text line. The variability in handwriting styles, diverse character representations, and alignment issues present unique challenges for OCR systems (Ivanova et al, 2023).

Although handwritten text presents challenges across AI tools, there are strategies to address this. For example, to improve recognition accuracy, ChatGPT uses probabilistic language models (such as n-grams or transformer-based models) to predict words based on context (Weisenthal. 2023). The authors also note that image distortion in real-world applications, such as blurring, pixelation, and artifacts, hinders OCR models’ accuracy. These factors introduce noise, causing OCR errors that disrupt LaTeX conversions. To address this, OCR models benefit from preprocessing techniques—such as denoising and contrast adjustments—to improve legibility and reduce distortion. Additionally, data augmentation methods that simulate image imperfections help OCR systems manage noisy images.

A promising solution for overcoming these OCR limitations involves incorporating advanced deep learning techniques, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), which can recognize complex mathematical structures more accurately. Additionally, introducing contextual embedding allows models to understand symbols based on surrounding syntax, enhancing differentiation between visually similar characters. Transformer models and hybrid use of CNNs have shown promise in improving OCR symbol recognition accuracy. Continuous evaluation of OCR performance through metrics like recall, precision, and F1 score, helping refine model capabilities iteratively through metrics like recall, precision, and F1 score help refine model capabilities iteratively as does the incorporation of active learning strategies to improve training efficiency (Orji et al, 2023). OCR models could also benefit by advancing mathematical content processing more precisely; this will serve to enhance model accuracy and support broader digitization efforts, improving accessibility to complex mathematical information for diverse audiences.

(CGPTv4) PROMPT: "I'm going to provide you with a JPG image of a document containing both typewritten and handwritten characters. Using the image I provide, I would like you to extract both the text and handwritten text from the following typewritten text on the image that I have included here in all caps as follows – The REG. NO., DATE OF BIRTH, DATE OF DEATH, AGE, CITY OR TOWN, RACE, SEX, MARRIED, NEVER MARRIED, WIDOWED, DIVORCED, OCCUPATION, CAUSE OF DEATH IMMEDIATE CAUSE and AS A CONSEQUENCE OF. Ignore any other personally identifiable information (PII) like the name of the decedent. When you extract this information, please organize it in spreadsheet form using the same headings. The image document is attached." (ATTACHED IMAGE BELOW)



MARYLAND STATE DEPARTMENT OF HEALTH
DIVISION OF VITAL RECORDS, 301 W. PRESTON STREET, BALTIMORE, MARYLAND 2120[redacted]
CERTIFICATE OF DEATH

1. DECEASED NAME (Type or print) First Middle Last [redacted]
2a. DATE OF DEATH JANUARY 13, 1974 2b. HOUR 9A M
3. SEX MALE 4. RACE White 5. DATE OF BIRTH [redacted] 6. AGE (in years last birthday) 82 YRS.
7a. BIRTHPLACE (State or foreign country) Maryland 7b. CITIZEN OF WHAT COUNTRY? United States 8. MARRIED ☒ NEVER MARRIED ☐ WIDOWED ☐ DIVORCED ☐ 9. COUNTY OF DEATH Wicomico
10. CITY OR TOWN OF DEATH Salisbury, Md. 11. NAME OF HOSPITAL OR INSTITUTION Peninsula General Hosp. 12a. USUAL OCCUPATION Salesman 12b. KIND OF BUSINESS OR INDUSTRY Industrial Prod.
13a. USUAL RESIDENCE STATE Maryland 13b. COUNTY Wicomico 13c. CITY OR TOWN Salisbury 13d. INSIDE CITY LIMITS? YES ☐ NO ☐ 13e. STREET AND NUMBER 1035 Riverside Drive
14. FATHER'S NAME First Middle Last [redacted] 15. MOTHER'S MAIDEN NAME First [redacted] Middle Last [redacted]
16a. WAS DECEASED EVER IN U.S. ARMED FORCES? No 16b. SOCIAL SECURITY NO. 17. INFORMANT (Wife) [redacted] Address same as 13e
18. CAUSE OF DEATH PART I. DEATH WAS CAUSED BY: IMMEDIATE CAUSE (a) DUE TO, OR AS A CONSEQUENCE OF (b) DUE TO, OR AS A CONSEQUENCE OF (c)
PART 2. OTHER SIGNIFICANT CONDITIONS CONTRIBUTING TO DEATH BUT NOT RELATED TO THE TERMINAL DISEASE CONDITION GIVEN IN PART I(a)
19a. DATE OF OPERATION 12-18-73 19b. CONDITION FOR WHICH OPERATION WAS PERFORMED 20a. AUTOPSY? YES ☐ NO ☒ 20b. IF YES, WERE FINDINGS CONSIDERED IN DETERMINING CAUSE OF DEATH?
21a. ACCIDENT, SUICIDE, HOMICIDE, OR UNDETERMINED 21b. TIME OF INJURY 21c. HOW INJURY OCCURRED 21d. INJURY OCCURRED 21e. PLACE OF INJURY 21f. LOCATION
22a. I certify that (I) (this hospital) attended the deceased from 12-17 1973 to 1-13 1974, that (I) (we) last saw the deceased alive on 1-12 1974 and that in (my) (our) opinion death occurred on the date and hour and from the causes stated above. (I) (we) (did) (did not) view the body after death.
22b. SIGNATURE 22c. DEGREE OR TITLE M.D. 22e. DATE SIGNED 1-13-74
22f. PHYSICIAN'S NAME (Type) 22g. ADDRESS Medical Center Salisbury
23a. BURIAL, CREMATION, REMOVAL Burial 23b. DATE 1/16/74 23c. NAME OF CEMETERY OR CREMATORY Greenmount Cemetery 23d. LOCATION (City or Town) Baltimore (County) (State) Maryland
24. FUNERAL DIRECTOR HOLLOWAY FUNERAL HOME, Salisbury, Maryland ADDRESS JAN 17 1974

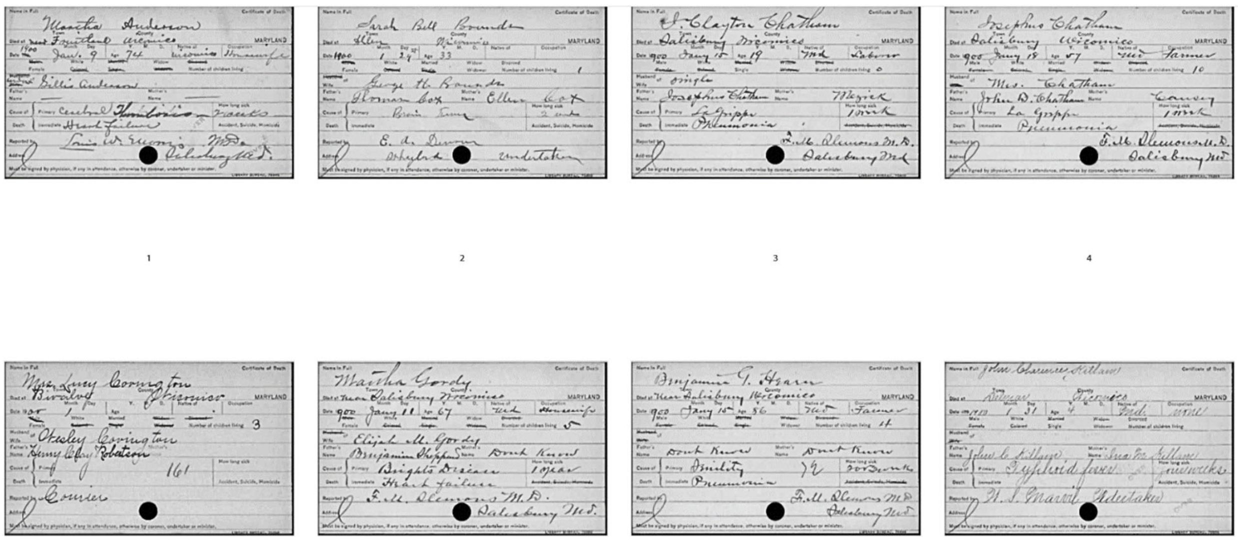
ChatGPT 4.0
Extracted Information

(CGPTv4) ERROR RESPONSE: "*I have extracted the text relevant to your request and displayed it in a structured form. However, there seem to be some issues with the OCR (Optical Character Recognition) accuracy due to the quality of the text. Some sections were not captured accurately, and I will need to adjust or refine the extraction method to capture it more precisely.*"

REG. NO.	DATE OF BIRTH	DATE OF DEATH	AGE

Figure 5a ChatGPT ATTEMPT 1

(CGPTv4) PROMPT: "Extract the handwritten text from the following typewritten text on the image that I have included here in all caps as follows – The DIED AT TOWN and COUNTY, DATE, MONTH, DAY, AGE, OCCUPATION, PRIMARY and IMMEDIATE CAUSE OF DEATH and HOW LONG SICK. Ignore any other personally identifiable information (PII) like the name of the decedent. When you extract this information, please organize it in spreadsheet form using the same headings. The document is attached." (ATTACHED IMAGE BELOW)

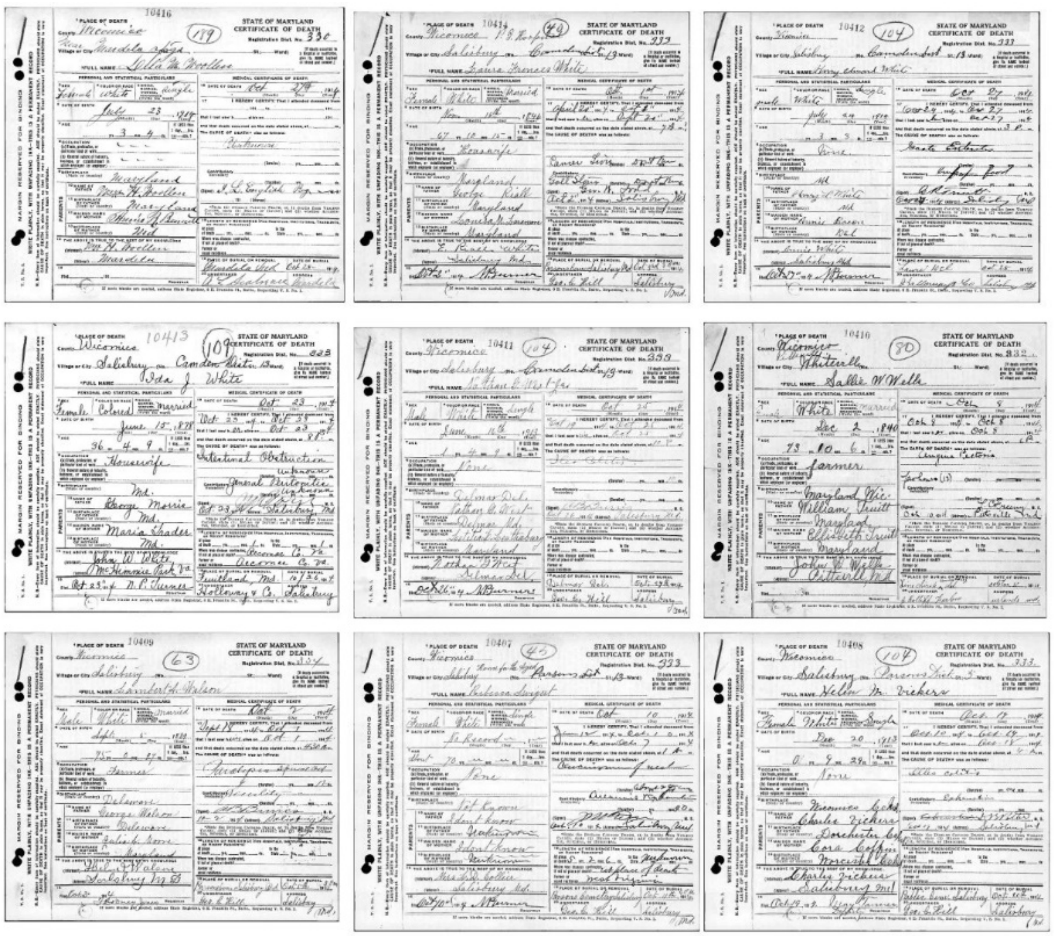


ChatGPT 4.0
Extracted Information

Died at Town	County	Date	Month	Day	Age	Occupation	Primary Cause of Death	Immediate Cause of Death	How Long Sick
Fruitland	Wicomico	1900	January	9	74	Housewife	Cerebral Thrombosis	Heart Failure	Immediate
Allen	Wicomico	1900	January	24	33		Brain Tumor		Immediate
Salisbury	Wicomico	1900	January	5	49	Laborer	La Grippe	Pneumonia	Immediate
Datcher	Wicomico	1900	January	7	47	Farmer	La Grippe	Pneumonia	Immediate
Bivalve	Wicomico	1900							161
Salisbury	Wicomico	1900	January	14	67	Housewife	Bright's Disease	Heart Failure	Immediate
Near Salisbury	Wicomico	1900	January	15	56	Farmer	La Grippe	Pneumonia	Immediate
Near Allen	Wicomico	1900	December	29	4		Syphilitic Fever	Pneumonia	Immediate

Figure 5b ChatGPT ATTEMPT 2

RESEARCHER PROMPT: *"Using the image I provide, I would like you to extract the handwritten text from the following typewritten text on the image that I have included here in all caps as follows – The DATE OF BIRTH (MONTH AND YEAR ONLY), DATE OF DEATH (MONTH AND YEAR ONLY), AGE, VILLAGE OR CITY, COLOR OR RACE, SEX, whether the decedent is SINGLE, MARRIED, WIDOWED, DIVORCED, OCCUPATION, CAUSE OF DEATH and DURATION; also include the CONTRIBUTORY (CAUSE OF DEATH) if included. When you extract this information, please combine it into one file and organize it in a spreadsheet form using the same headings provided. The document is attached."* (ATTACHED IMAGE BELOW)



ChatGPT 4.0
Extracted Information

Date of Birth (Month and Year)	Date of Death (Month and Year)	Age
June 1887	July 1945	58
Jan 1925	Feb 1956	31
Feb 1899	Mar 1962	63

Figure 5c ChatGPT ATTEMPT 3

RESEARCHER PROMPT: *""I'm going to provide you with documents containing both typewritten and handwritten characters. Using the JPEG images, I would like you to extract the handwritten text from the following typewritten text on the image that I have included here in all caps as follows – The DATE OF BIRTH (MONTH AND YEAR ONLY), DATE OF DEATH (MONTH AND YEAR ONLY), AGE, VILLAGE OR CITY, COLOR OR RACE, SEX, whether the decedent is SINGLE, MARRIED, WIDOWED, DIVORCED, OCCUPATION, CAUSE OF DEATH and DURATION; also include the CONTRIBUTORY (CAUSE OF DEATH) if it included. Ignore any other personally identifiable information (PII) like the name of the decedent, father, mother, spouse, etc. Please only use the information from the documents provided and do not include any extraneous details or your personal ideas or interpretations. When you extract this information, please number each entry sequentially and organize it in spreadsheet form using the same headings in the specific order I have provided. The documents are attached."*
(ATTACHED IMAGE BELOW)

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_35

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_36

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_37

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_38

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_39

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_40

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_41

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_42

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_43

1728250240439-997dec2e-4088-4f61-9854-70fed59f573c_44

	A	B	C	D	E	F	G	H	I	J	K
1	Date of Birth (Month and Year)	Date of Death (Month and Year)	Age	Village or City	Color or Race	Sex	Marital Status	Occupation	Cause of Death	Duration	Contributory (Cause of Death)
2	June 1887	July 1945	58	Salisbury	White	Male	Married	Farmer	Heart Attack	2 days	None
3	Jan 1925	Feb 1956	31	Camden	White	Female	Single	Housekeeper	Pneumonia	1 month	Diabetes
4	Feb 1899	Mar 1962	63	Wicomico	White	Male	Widowed	Carpenter	Kidney Failure	3 months	Hypertension
5											
6	(3 / 10 processed)										
7											

Figure 5d ChatGPT ATTEMPT 4 – Folder link, output to spreadsheet)

Analysis of Manual Data Entry

For this study, analysis was focused on chronic diseases associated with $PM_{2.5}$ air pollution, specifically cardiovascular disease, hypertension (high blood pressure), asthma, cancer, and diabetes. Other causes of death prevalent during the study period, but not included in this analysis, range from asthma to tuberculosis, reflecting the respiratory and infectious disease burdens; additionally, there were noticeable trends in mortality due to malnourishment, pneumonia, acute syphilis, dysentery, stillbirth, kidney disease and age-related/attributed demise.

Observation years (1911 to 1914) of the dataset include key variables such as cause of death, year of death, age at death, duration of illness, race, gender, marital status, and occupation. Referring to the results shown summarized in Table 3, the mean age at death averages 52.64 years (SD = 21.49), with a median of 56, reflecting a moderately spread distribution skewed slightly left (-0.35) (Table 3).

Race is dichotomous containing only Black and White decedents in this sample, with a mean of 1.70 (SD = 0.46) (Table 3). In this sample, White decedents show a higher incidence of mortality from diabetes, whereas Black decedents are more represented in heart disease deaths.

Deaths by gender (Figure 5a) has a mean of 1.49 (SD = 0.50), with values close to the midpoint between both groups (i.e., male and female), suggesting nearly equal representation. The deaths by marital status of decedents (mean = 2.69, SD = 0.87) exhibits moderate variability and a slight positive skew (0.56), with most responses clustering around the median of 2. Occupation years span a considerable range (1–22), averaging 13.92 years (SD = 5.24), with a slight negative skew (-0.63), reflecting a tendency toward shorter work durations. The duration of illness averages 1.52 years (SD = 2.23), with significant skewness (2.55) indicating that most durations are short, as evidenced by a median of 1. Finally, the cause of death has a mean of 4.50 (SD = 1.27) and a slightly negatively skewed distribution (-0.40), with values ranging from 1 to 6. Analysis reveals some disparities in mortality specifically from cardiovascular disease, diabetes, and hypertension. While these findings align with existing literature on health disparities, the absence of direct $PM_{2.5}$ exposure data precludes definitive conclusions regarding causality.

Table 3 – Summary of Manual Data Entry, (Kelly, 2025)

Category	Key Findings	Statistical Details	Skewness & Distribution
Primary Causes of Death	Cardiovascular disease, hypertension, asthma, cancer, diabetes, malnourishment, pneumonia, syphilis, dysentery, stillbirth, kidney disease, age-related mortality.	Mean = 4.50, SD = 1.27	Slight negative skew (-0.40), certain causes more frequent.
Observation Period	1911–1914 dataset includes cause of death, year, age, illness duration, race, gender, marital status, occupation.	—	—
Age at Death	Mean age at death was 52.64 years, with a median of 56.	Mean = 52.64, SD = 21.49, Median = 56	Slight negative skew (-0.35), moderately spread.
Race Distribution	Only Black and White individuals included. White decedents had higher diabetes mortality; Black decedents show more heart disease deaths.	Mean = 1.70, SD = 0.46	Negative skew (-0.84), slight overrepresentation of lower-coded race.
Gender Distribution	Nearly equal representation of males and females.	Mean = 1.49, SD = 0.50	Balanced representation.
Marital Status	Moderate variability, clustering around median of 2.	Mean = 2.69, SD = 0.87, Median = 2	Slight positive skew (0.56).
Occupation Years	Work duration ranged from 1–22 years, with an average of 13.92 years.	Mean = 13.92, SD = 5.24	Slight negative skew (-0.63), tendency for shorter careers.
Duration of Illness	Most illnesses were short, with a median of 1 year.	Mean = 1.52 years, SD = 2.23, Median = 1	Highly skewed (2.55), indicating most cases were brief.
Health Disparities	Mortality disparities are noted in cardiovascular disease, diabetes, hypertension.	—	Limited by lack of granular racial data.
Environmental Factors	Chronic diseases linked to PM2.5 exposure but no direct pollution data available.	—	Causal conclusions cannot be drawn.

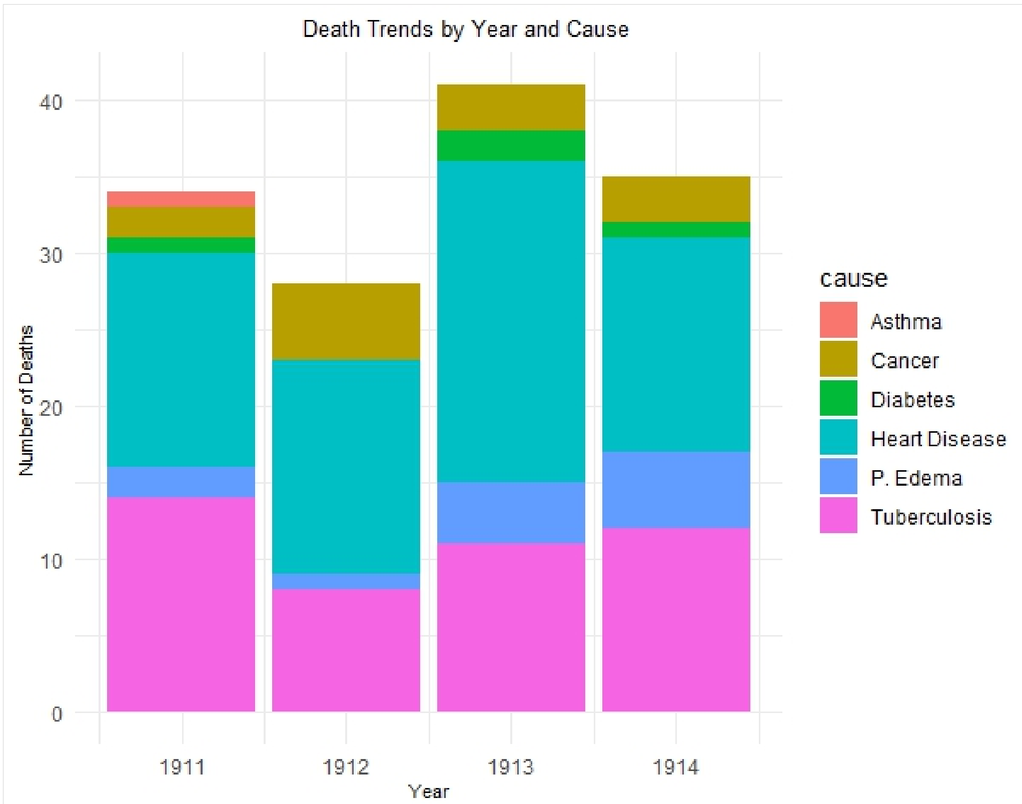


Figure 6a Preliminary Analysis – Death Trends by Year and Cause

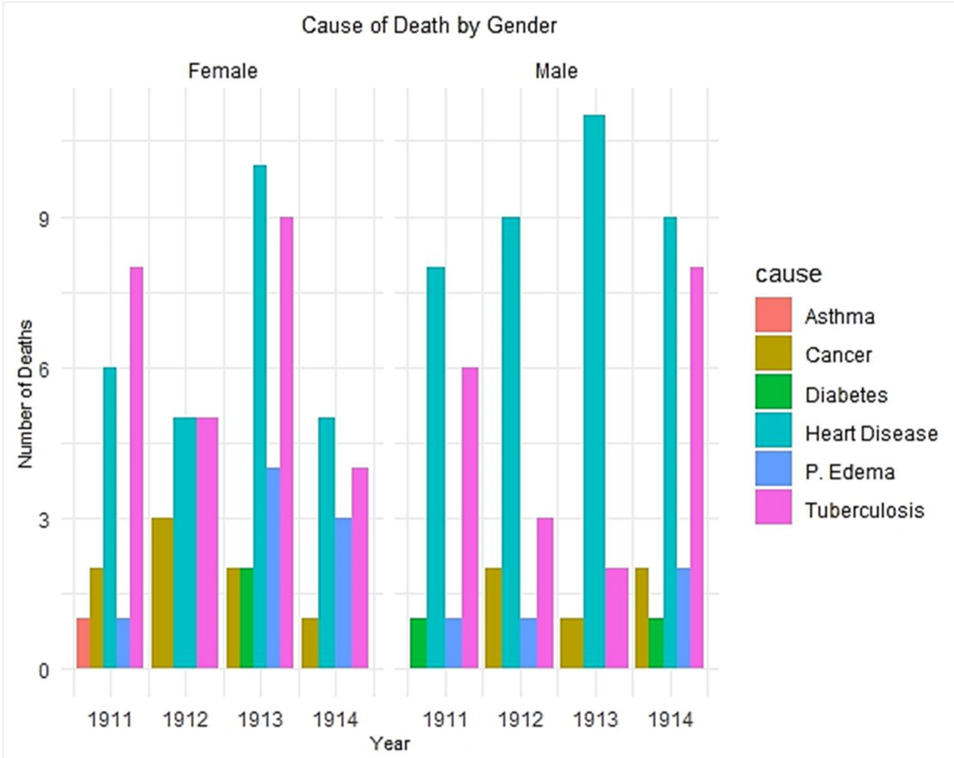


Figure 6b Preliminary Analysis – Cause of Death by Gender

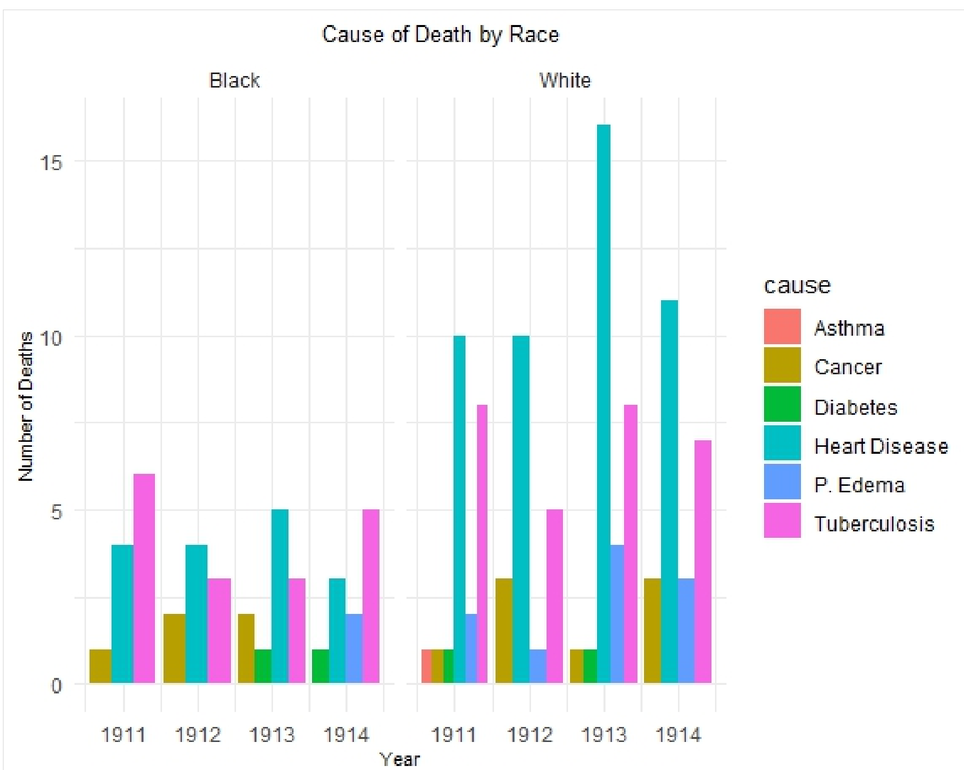


Figure 6c Preliminary Analysis – Cause of Death by Race

There are nominal variations in age, work duration, and cause of death distributions across the observation period. For example, the mean age in 1914 (58.69 years) is notably higher than in previous years, possibly indicating increased longevity or survivorship biases. Conversely, work duration exhibits significant fluctuation, with the highest mean in 1911 (25.18 years) and the lowest in 1912 (6.00 years). These temporal differences warrant longitudinal analyses to identify trends linked to infrastructural and policy and zoning changes, emerging industries and technologies, and advances in healthcare practice and accessibility.

Strengths of the Study

The general aim of this research was to measure the relationship between race, traffic-related $PM_{2.5}$, and its health effects on communities living near major roadways. Two distinct objectives of this project were: (1) to collect and compile missing chronic disease mortality data typically associated with prolonged exposure to $PM_{2.5}$ environmental health data for the period from 1911 to 1922, and (2) to determine if AI tools can effectively extract historical data from archived death certificates from the target period (i.e., between 1911 to 1922) .

The AI approaches used in this study, although exploratory, yielded sufficient data to validate its potential for handwriting extraction and dataset development; the potential compelling opportunity presented here is an initial framework for the application of evolving AI technologies to collect more comprehensive and multifaceted environmental health data, innovating novel approaches to interdisciplinary research and interventions.

Overall, the results of this preliminary analysis do provide some insight into the characteristics of the population sample from Georgetown, Salisbury, Maryland, during the early 20th century and early drivers of mortality.

The data collected reflects the demographic, occupational, and mortality attributes across four consecutive years (1911-1914), which could provide a viable basis for comparison with later period studies of the same target area for similar mortality trends.

Further analysis of potential correlations between environmental factors, such as $PM_{2.5}$ exposure, and chronic disease prevalence with combined with demographic variables is required to identify any direct correlations.

Limitations of this study

While this scholarship highlights the potential viability for using AI tools to extract data from archival sources (e.g., images of death certificates), it also reveals the complexity of obstacles to generating useful datasets, such as the high variability of image quality and features, handwriting and text, the complexity of algorithmic mathematics (Bohn, et al, 2024) required for model training, the lack of availability and the granularity of environmental sample data (e.g., soil core samples, local air quality, etc.). Each of these would be needed to combine with other sources such as census and health data to provide for more holistic analysis.

In addition to the challenges presented by image complexity, there are several other factors that limit the findings of this work. For instance, there is a need for a longer data period to examine potential trends for chronic disease-related deaths but also $PM_{2.5}$ concentration trends by zip code. the absence of localized $PM_{2.5}$ air quality data for the case study area requires that estimates be extrapolated from the nearest air quality monitoring station located on the site of Salisbury Regional Airport (SBY) at 5485 Airport Terminal Rd, Salisbury, MD 21804 (IQAir/SBY, n.d.). While this could be a reliable approach, the roughly six-mile distance of the monitoring station from the Georgetown community may not accurately account for residential exposures experienced adjacent to roadways, specifically US Routes 13 and 50 (Walsh, Shewell, 2002).

Another consideration is the lack of temporal alignment between exposure periods and disease outcomes. Prior to the Clean Air Act of 1970, there was no governmental oversight for air quality and further, there was no governing body for the state of Maryland that was collecting health and disease data until the early 1980s. These obvious gaps, while motivators for the study are inherently confounding aspects of the research findings that precluded any comprehensive interaction or time-series analyses.

Improved preprocessing and prompt engineering could be used with the remaining original archival documents collected - 1911 to 1922 (n = 1680) and 1973 to 1979 (n = 1173) to produce more accurate results; the larger sample size presents an expanded opportunity for testing, training, and analysis.

It was beyond the scope of this research to provide analysis of environmental quality data for the case study area; however, the potential utility of sampling the air, soil, and water conditions in the target area cannot be understated. For example, the installation of air quality monitoring units at strategic points within the case study area would provide a more accurate reading of ground truth $PM_{2.5}$ concentration data.

CHAPTER 4: CONCLUSION

Broader Impacts

AI's role in healthcare extends beyond technological advancement; it necessitates an ecosystem that values data integrity, equitable patient outcomes, and ethical responsibility. While current research illustrates the capacity of AI to reshape public health surveillance and management, there is an added emphasis on the importance of methodological rigor and ethical frameworks.

This research makes contributions toward understanding the drivers of chronic disease especially in minoritized communities that continue to pose complex and dynamic challenges. Making equitable strides in minority public health outcomes means considering not only the genetic precursors of epidemiology but also the myriad social and historical factors that have contributed to chronic disease morbidity and mortality in historically marginalized communities. The development of comprehensive analytical models to collect the data needed to address gaps in our knowledge about minority health risk will continue to be relevant for all communities managing chronic disease. This study establishes a basis for a standardized selection and classification model of data collection and a foundational basis for advancing innovations in epigenetic research, informing stakeholder policy, and funding decisions, and improving community public health equity and intervention, locally and globally.

Future Research

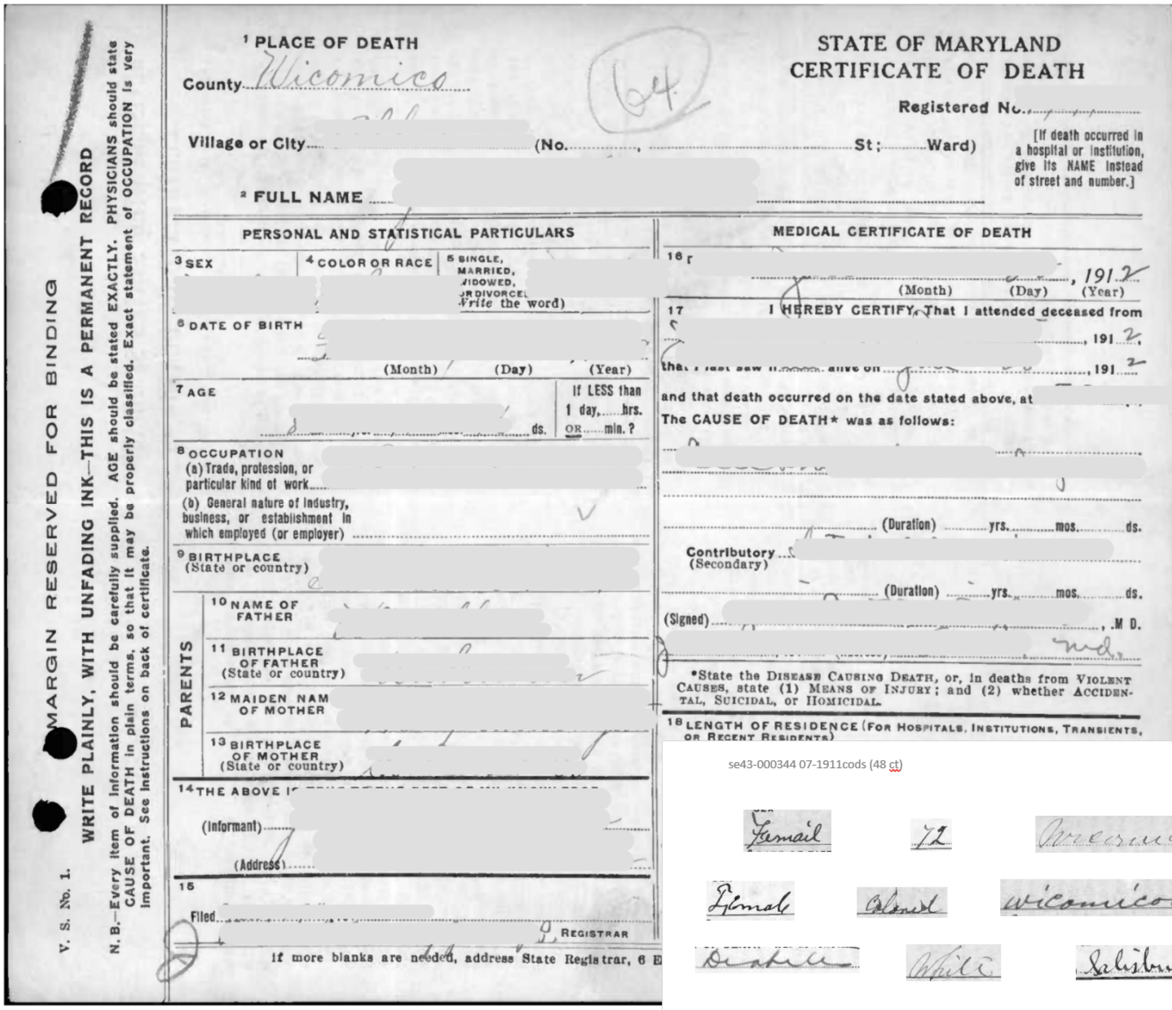
In a future study, *Exposomic* insights could be improved by combining available data from multiple state repositories, such as traffic volume data by county from the Maryland Department of Transportation, chronic disease data from the Maryland Department of Health, and Census data from the United States Census Bureau; sing GIS to geocode concentrations and exposures to $PM_{2.5}$ air pollution from vehicle emissions would also be especially relevant given the increasing traffic volume of US Routes 13 and 50. Using these combined datasets, tabular data could be adjoined, and predictive analyses conducted to compare and contrast potentially correlated variables such as $PM_{2.5}$ concentrations, traffic volume trends and chronic disease mortality, exploring how these are represented in the datasets from AI extracted datasets. By joining these data sources, a broader spectrum of exposures impacting the CHANS can be better represented for analysis (Boussalis, et al., 2015).AI has emerged as a pivotal technology that continues to evolve, driven by advancements in computational power and data availability. Its diverse applications across many domains, particularly in environmental health, underscore the importance of integrating AI methodologies to enhance our understanding of complex systems and improve public health outcomes. The significant potential of artificial intelligence (AI) to enhance environmental health surveillance and management. Through various approaches in data analysis, predictive modeling, and pragmatic research applications is vast and rapidly expanding. By synthesizing the work of multiple studies, we observe AI's transformative potential in healthcare, while also identifying critical ethical and methodological concerns that must be addressed to ensure effective, equitable, and transparent deployment

The evolving landscape of environmental science requires embracing interdisciplinary frameworks like the CHANS to address pressing public health and environmental challenges. By acknowledging the reciprocal nature of human and environmental systems, this research aims to contribute to sustainable policy development and improve quality of life for vulnerable communities. The purpose of viewing environmental and public health within the context of the CHANS is to facilitate better knowledge integration across disciplines and to improve human-environment interaction modeling, capturing the feedback systems, externalities, and emergent properties that comprise the environmental health equation. It emphasizes creating conceptual frameworks and shared models to facilitate communication among stakeholders, practitioners, and researchers from diverse fields. To develop comprehensive environmental health models crucial for understanding complex systems and guiding public policy, the CHANS framework combined with emerging AI technologies presents an unprecedented opportunity to advance research understanding and quality of life.

APPENDICES

Appendix A

Redacted Certificate of Death and Handwriting Sample (Maryland State Archives)



V. S. No. 1.

MARGIN RESERVED FOR BINDING

WRITE PLAINLY, WITH UNFADING INK—THIS IS A PERMANENT RECORD

N. B.—Every item of information should be carefully supplied. AGE should be stated EXACTLY. PHYSICIANS should state CAUSE OF DEATH in plain terms, so that it may be properly classified. Exact statement of OCCUPATION is very important. See instructions on back of certificate.

STATE OF MARYLAND
CERTIFICATE OF DEATH

1 PLACE OF DEATH

County Wicomico

64

Registered No.

Village or City [redacted] (No. , St; Ward) [If death occurred in a hospital or institution, give its NAME instead of street and number.]

2 FULL NAME [redacted]

PERSONAL AND STATISTICAL PARTICULARS

3 SEX [redacted] 4 COLOR OR RACE [redacted] 5 SINGLE, MARRIED, WIDOWED, OR DIVORCED (Write the word) [redacted]

6 DATE OF BIRTH [redacted] (Month) (Day) (Year)

7 AGE [redacted] ds. If LESS than 1 day, hrs. OR min.?

8 OCCUPATION (a) Trade, profession, or particular kind of work [redacted] (b) General nature of industry, business, or establishment in which employed (or employer)

9 BIRTHPLACE (State or country) [redacted]

PARENTS

10 NAME OF FATHER [redacted]

11 BIRTHPLACE OF FATHER (State or country) [redacted]

12 MAIDEN NAME OF MOTHER [redacted]

13 BIRTHPLACE OF MOTHER (State or country) [redacted]

14 THE ABOVE IS TRUE TO THE BEST OF MY KNOWLEDGE

(Informant) [redacted]

(Address) [redacted]

15 Filed [redacted] REGISTRAR

MEDICAL CERTIFICATE OF DEATH

16 [redacted], 1912 (Month) (Day) (Year)

17 I HEREBY CERTIFY, That I attended deceased from [redacted], 1912

that I last saw h alive on [redacted], 1912

and that death occurred on the date stated above, at [redacted]

The CAUSE OF DEATH* was as follows:

[redacted]

(Duration) yrs. mos. ds.

Contributory (Secondary) [redacted]

(Duration) yrs. mos. ds.

(Signed) [redacted], M. D.

[redacted] Md.

*State the DISEASE CAUSING DEATH, or, in deaths from VIOLENT CAUSES, state (1) MEANS OF INJURY; and (2) whether ACCIDENTAL, SUICIDAL, or HOMICIDAL.

18 LENGTH OF RESIDENCE (FOR HOSPITALS, INSTITUTIONS, TRANSIENTS, OR RECENT RESIDENTS)

If more blanks are needed, address State Registrar, 6 E

Appendix B

NAAQS Table

Units of measure for the standards are parts per million (ppm) by volume, parts per billion (ppb) by volume, and micrograms per cubic meter of air ($\mu g/m^3$).

Pollutant [links to historical tables of NAAQS reviews]		Primary/ Secondary	Averaging Time	Level	Form
Carbon Monoxide (CO)		primary	8 hours	9 ppm	Not to be exceeded more than once per year
			1 hour	35 ppm	
Lead (Pb)		primary and secondary	Rolling 3-month average	0.15 $\mu g/m^3$ (1)	Not to be exceeded
Nitrogen Dioxide (NO_2)		primary	1 hour	100 ppb	98th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		primary and secondary	1 year	53 ppb (2)	Annual Mean
Ozone (O_3)		primary and secondary	8 hours	0.070 ppm (3)	Annual fourth-highest daily maximum 8-hour concentration, averaged over 3 years
Particle Pollution (PM)	$PM_{2.5}$	primary	1 year	9.0 $\mu g/m^3$	annual mean, averaged over 3 years
		secondary	1 year	15.0 $\mu g/m^3$	annual mean, averaged over 3 years
		primary and secondary	24 hours	35 $\mu g/m^3$	98th percentile, averaged over 3 years
	PM_{10}	primary and secondary	24 hours	150 $\mu g/m^3$	Not to be exceeded more than once per year on average over 3 years
Sulfur Dioxide (SO_2)		primary	1 hour	75 ppb (4)	99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		secondary	1 year	10 ppb	annual mean, averaged over 3 years

(1) In areas designated nonattainment for the Pb standards prior to the promulgation of the current (2008) standards, and for which implementation plans to attain or maintain the current (2008) standards have not been submitted and approved, the previous standards (1.5 µg/m3 as a calendar quarter average) also remain in effect.

(2) The level of the annual NO_2 standard is 0.053 ppm. It is shown here in terms of ppb for the purposes of clearer comparison to the 1-hour standard level.

(3) Final rule signed October 1, 2015, and effective December 28, 2015. The previous (2008) O_3 standards are not revoked and remain in effect for designated areas. Additionally, some areas may have certain continuing implementation obligations under the prior revoked 1-hour (1979) and 8-hour (1997) O_3 standards.

(4) The previous SO_2 standards (0.14 ppm 24-hour and 0.03 ppm annual) will additionally remain in effect in certain areas: (1) any area for which it is not yet 1 year since the effective date of designation under the current (2010) standards, and (2)any area for which an implementation plan providing for attainment of the current (2010) standard has not been submitted and approved and which is designated nonattainment under the previous SO_2 standards or is not meeting the requirements of a SIP call under the previous SO_2 standards (40 CFR 50.4(3)). A SIP call is an EPA action requiring a state to resubmit all or part of its State Implementation Plan to demonstrate attainment of the required NAAQS.

Appendix C

NLP Example - ChatGPT Query Output

FkdwJSW#fdq#eh#xvhg#iru#wh{w#dqdo|vlv#lq#grfxphqw#uhfrjqlwlrq#e|#ohyhudjlqj#lwv#qdwxudo#odqjxdjh#surfhvvlqj#+QOS,#fdsdelolwlhv#wr#h{wudfw/#fodvvli|/#dqg#lqwhusuhw#wh{wxdo#lqirupdwlrq#iurp#pxowlsoh#w|shv#ri#grfxphqwv1#Khuh#lv#krz#lw#fdq#eh#dssolhg=#

41 Rswlfdo#Fkdudfwhu#Uhfrjqlwlrq#+RFU,#Hqkdqfhphqw#
  - FkdwJSW#fdq#surfhvv#wh{w#h{wudfwhg#iurp#lpdjhv#ru#vfdqqhg#grfxphqwv#xvlqj#RFU#wrrov#olnh#Whvvhudfw#ru#Jrrjoh#Ylvlrq#DL1#
  - Lw#fdq#khos#fohdq/#fruuhfw/#dqg#vwuxfwxuh#RFU0h{wudfwhg#wh{w#e|#uhprylqj#qrlvh/#il{lqj#huuruv/#dqg#lpsurylqj#uhdgdelolw|1#

51 Qdphg#Hqwlw|#Uhfrjqlwlrq#+QHU,#
  - FkdwJSW#fdq#lghqwli|#dqg#fdwhjrul}h#lpsruwdqw#hqwlwlhv#vxfk#dv#qdphv/#gdwhv/#rujdql}dwlrqv/#dqg#orfdwlrqv#zlwklq#d#grfxphqw1#
  - Xvhixo#iru#lqgh{lqj#ohjdo#grfxphqwv/#lqyrlfhv/#frqwudfwv/#dqg#dfdghplf#sdshuv1#

61 Grfxphqw#Fodvvlilfdwlrq#dqg#Fdwhjrul}dwlrq#
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  - Lw#fdq#dovr#dsso|#wrslf#prgholqj#whfkqltxhv#wr#fdwhjrul}h#odujh#grfxphqw#uhsrvlwrulhv1#

71 Nh|zrug#H{wudfwlrq#dqg#Vxppdul}dwlrq#
  - E|#dqdo|}lqj#wkh#wh{w/#FkdwJSW#fdq#h{wudfw#nh|#whupv#dqg#skudvhv#wr#vxppdul}h#grfxphqwv#hiilflhqwo|1#
  - Xvhixo#iru#dxwrpdwhg#uhsruw#jhqhudwlrq/#h{hfxwlyh#vxppdulhv/#dqg#grfxphqw#lqgh{lqj1#

81 Vhqwlphqw#dqg#Wrqh#Dqdo|vlv#
  - FkdwJSW#fdq#dqdo|}h#wkh#vhqwlphqw#ru#wrqh#ri#d#grfxphqw/#ghwhfwlqj#zkhwkhu#lw#lv#irupdo/#ohjdo/#shuvxdvlyh/#ru#hprwlrqdo1#
  - Ehqhilfldo#iru#dqdo|}lqj#fxvwrphu#ihhgedfn/#ohjdo#glvsxwhv/#dqg#frusrudwh#frppxqlfdwlrqv1#

91 Sodjldulvp#Ghwhfwlrq#dqg#Vlplodulw|#Dqdo|vlv#
  - FkdwJSW#fdq#frpsduh#grfxphqwv#wr#ghwhfw#gxsolfdwhg#frqwhqw/#sdudskudvhg#wh{w/#ru#srwhqwldo#sodjldulvp1#
  - Khosixo#lq#dfdghplf#zulwlqj/#mrxuqdolvp/#dqg#ohjdo#frpsoldqfh1#

:1 Dxwrpdwhg#Gdwd#H{wudfwlrq#
  - H{wudfwlqj#vwuxfwxuhg#lqirupdwlrq#vxfk#dv#wdeohv/#qxphulfdo#ydoxhv/#ru#phwdgdwd#iurp#xqvwuxfwxuhg#grfxphqwv1#
  - Xvhixo#iru#ilqdqfldo#grfxphqw#surfhvvlqj/#phglfdo#uhfrugv/#dqg#exvlqhvv#lqwhooljhqfh1#

;1 Txhvwlrq#Dqvzhulqj#iurp#Grfxphqwv#
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<1 Wh{w#Wudqvodwlrq#dqg#Pxowlolqjxdo#Dqdo|vlv#
  - FkdwJSW#fdq#wudqvodwh#grfxphqwv#dqg#dqdo|}h#frqwhqw#dfurvv#gliihuhqw#odqjxdjhv1#
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431 Frqwh{wxdo#Vhdufk#dqg#Uhwulhydo#

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- Xvhixo#iru#odujh#grfxphqw#gdwdedvhv/#ohjdo#dufklyhv/#dqg#uhvhdufk#uhsrvlwrulhv1#

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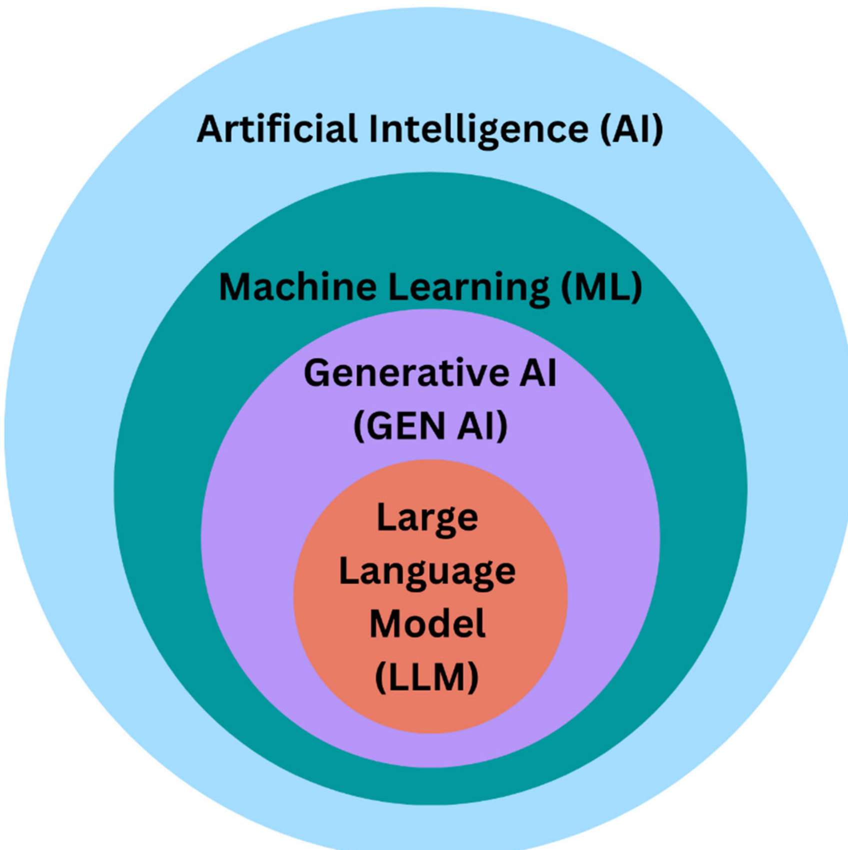
dqg#nqrzohgjh#h{wudfwlrq/#lpsurylqj#hiilflhqf|#lq#vhfwruv#olnh#ohjdo/#ilqdqfh/#khdowkfduh/#uhvhdufk/#dqg#
hqwhusulvh#pdqdjhphqw1##

Appendix D

Understanding Artificial Intelligence Hierarchy: How AI, ML, Gen AI, and LLM Are Related

(Yousry, 2025)



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Appendix E

Sanborn Fire Insurance Map from Salisbury, Maryland - Humphrey's Pond, Library of Congress

Newspaper article - Baltimore Sun, May 29, 1909 "Salisbury Dam Break", gclock.com

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SCENE OF DISASTER IN SALISBURY

HARBOR AT SALISBURY AND LAKE HUMPHREYS

FLOOD IN SALISBURY

Dam Breast Of Lake Humphreys Gives Way With A Crash.

CAMDEN BRIDGE WASHED OUT

Part Of Division Street Torn Away, Gas Mains And Sewer Pipes Are Torn Out And Other Damage Done

[Special Dispatch to the Baltimore Sun.]

Salisbury, Md., May 28.—The dam breast of Lake Humphreys gave way with a crash today, destroying bridges and waterfront property, completely tearing away over 200 feet of Division street and undermining Camden avenue in the heart of the city, completely cutting off two sections of the city; wrecking sail vessels, launches, demolishing fishing outfits and flooding the lowlands, farms and truck land along the Wicomico river.

Lake Humphreys is situated in the center of the city and covers about 30 acres. The break was at its head gateway, which abuts on South Division street. The crash, that accompanied the giving way of the break sounded throughout the city, and the torrents tore through the Humphreys estate, uprooting trees, carrying away telephone and electric light poles, bursting the gas mains and city sewer pipes to Camden avenue bridge, where it carried away the abutments of the bridge and undermined the street approaches.

When the water reached the harbor it played havoc, tearing large sailing vessels from their moorings, causing them to ram the wharves, and jam smaller craft. The beautiful houseboat Idylwilde, owned, by Captain Walley, which was lying in the harbor, was badly wrecked. Captain Leatherbury also had his yacht wrecked.

Other boat, owners were caused considerable loss. Traffic has been completely shut off between the northern and southern parts of the city, the course of the flood being through the central part of the city. Farmers and people from the outlying districts cannot reach the business section until temporary bridges are constructed.

The city will be the heaviest loser, its loss amounting to about $7,000. The gas, telephone and electric light companies will also suffer much loss. The Humphreys estate will be the largest individual loser.

The cause of the breaking of the dam is attributed to the recent heavy rains. Mr. R. P. Bailey and Wood Harmon, who were crossing the bridgeway about the time of the break, narrowly escaped with their lives. They got out of way of the flood and then notified other pedestrians, who sought places of safety. Crowds of people soon assembled, and under the charge of Chief of Police Disharoon the crowds were driven back from the dangerous points and the break roped off.

Tonight the harbor is full of floating timber, stumps, trees, etc. The city authorities will immediately begin the erection of two temporary bridges, which will be completed in a few days, when traffic will be resumed. The southern part of the city, with a population of about 3,000, will be in darkness and without telephonic connection until the bridges can be constructed.

DR. MERRITT'S BAIL STANDS

Talbot Court Refuses To Cut It Down From $5,000 To $2,500.

[Special Dispatch to the Baltimore Sun.]

Easton, Md., May 28.—The Court today refused to reduce the amount of the bail bond of Dr. James B. Merritt, accused of criminal malpractice, by which a young girl lost her life last fall. Dr. Merritt was released on $5,000 bail and did not appear for trial when his case was called and the Court ordered his bail bond forfeited. Col. James C. Mullikin, his counsel, petitioned the Court to reduce the bond to $2,500.

Judge Pearce, who delivered the opinion of the court, said:

"It is not in the province of this court to reduce the bond, as that lies solely with

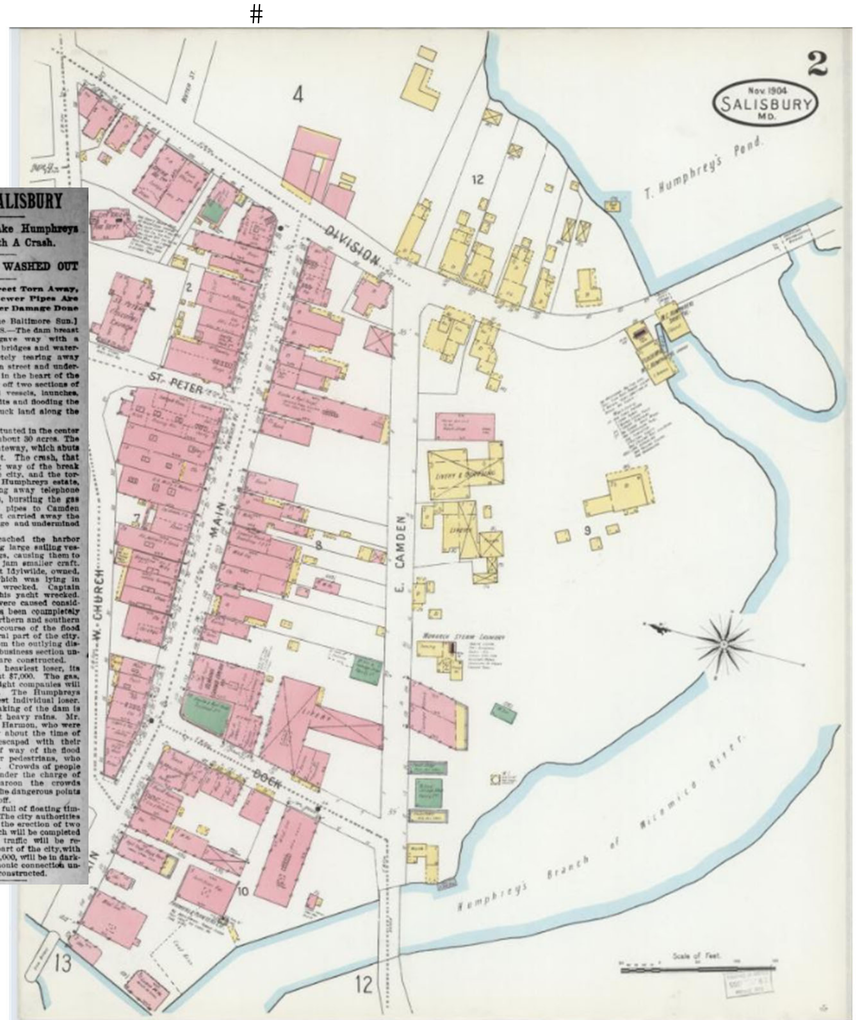
DIES AS RESULT OF FIGHT

Hiram Truitt Had Abdomen Slashed Open With A Razor.

[Special Dispatch to the Baltimore Sun.]

Salisbury, Md., May 28.—Hiram Truitt, 22 years old, son of James Truitt, a highly respected resident of the eastern section of Wicomico county, died at Powellville today as a result of injuries received in a fight last Saturday night. The affair had been kept very quiet by the principals and their friends, but on young Truitt becoming very ill today, the news of the trouble spread rapidly.

Truitt, Joseph Bodly and Edward Dennis had a dispute Saturday night which ended in a fight. After the fight the three went to the barber shop of James Donnaway and there again became involved in an



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